

Polynomial-Time MIMO Detection at the Maximum-Likelihood Threshold

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Abstract

We prove that exact block recovery in the square Gaussian binary MIMO model can be achieved in polynomial time at the same first-order SNR threshold as exhaustive maximum-likelihood detection. Specifically, for

$$\mathbf{y} = \sqrt{\rho/N} \mathbf{H} \mathbf{x}^* + \mathbf{w}, \quad \mathbf{x}^* \in \{\pm 1\}^N,$$

and independent standard Gaussian $\mathbf{H} \in \mathbb{R}^{N \times N}$ and $\mathbf{w}$, rounded linear MMSE followed by steepest single-bit descent recovers $\mathbf{x}^*$ with failure probability tending to zero, uniformly over every transmitted word and every $\rho \geq 2 \log N$, using $O(N^3)$ unit-cost exact-real arithmetic operations. To the best of our knowledge, no prior work gives polynomial-time exact block recovery at the ML boundary for this setting; the closest prior square-system theorem, for the box relaxation, has first-order threshold $4 \log N$. The proof places the rounded LMMSE estimate at sublinear Hamming distance from the truth, and then establishes, uniformly over every error set the local search can visit, that some wrong bit offers a quantified cost decrease while an objective barrier confines the search path. Conversely, if $\rho \leq 2 \log N - \log \log N - s_N$ with $s_N \rightarrow \infty$ and $s_N = o(\log N)$, then a one-bit neighbor beats the transmitted word with probability tending to one, so even ML detection fails. Therefore, the statistical and polynomial-time exact-recovery thresholds coincide to first order.

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1 Introduction

Binary MIMO detection is a random instance of binary least squares. The same objective arises in optimum multiuser detection for synchronous CDMA, and it is a finite-alphabet integer-least-squares problem closely related to closest-vector search on a lattice with a noisy planted point [Verdú, 1989; Fincke and Pohst, 1985; Agrell et al., 2002]. For arbitrary channel matrices and observations, optimum multiuser detection is NP-hard [Verdú, 1989]. A Gaussian channel is more structured, however, and worst-case hardness does not determine its typical complexity. This paper asks whether an explicit polynomial-time detector can achieve exact block recovery at the same first-order SNR threshold as exhaustive maximum-likelihood detection.

Early average-case studies of ML detection include Hassibi and Vikalo [2001, 2002, 2005]; Vikalo

and Hassibi [2005], who derived the expected complexity of the sphere decoder (Fincke–Pohst enumeration [Fincke and Pohst, 1985]), averaged over the channel and noise, and reported polynomial behavior over broad ranges of practically relevant dimensions and SNRs. However, Jaldén and Ottersten [2005] showed that this behavior does not persist in the growing-dimension limit: at every fixed SNR, the expected complexity is exponential in the dimension. A key geometric reason is that, to contain the transmitted point with nonvanishing probability, the search sphere must have squared radius proportional to the dimension.

The first-order SNR scaling for successful ML block recovery was identified by Hansen et al. [2009], and subsequent work [Hassibi et al., 2014, Lemma IV.2] states the condition explicitly as

$$\rho > 2 \log N + f(N), \quad f(N) \rightarrow \infty,$$

under which the ML block-error probability vanishes. At the threshold scale $\rho = \Theta(\log N)$, the primary sphere-decoding lower bound specializes in the present normalization to

$$C(N) \geq 2^{N/(4\rho+2)} - 1 = \exp\{\Theta(N/\log N)\},$$

which is superpolynomial but subexponential in N [Jaldén and Ottersten, 2005, Eq. (29)]. Under only $\rho = O(\log N)$, this lower bound is at least $\exp\{\Omega(N/\log N)\}$ and can be exponential when $\rho = O(1)$. These works established an ML achievability result but did not prove a matching converse. A direct calculation involving the N one-bit neighbors predicts the same first-order constant. If $Q(t) = \Pr\{Z \geq t\}$, $Z \sim \mathcal{N}(0, 1)$, then a given neighbor beats the transmitted word with probability asymptotic to $Q(\sqrt{\rho})$ in the relevant regime. The expected number of such neighbors is therefore approximately $NQ(\sqrt{\rho})$, and the transition $NQ(\sqrt{\rho}) \asymp 1$ occurs at

$$\rho = 2 \log N - \log \log N + O(1).$$

Our constructive theorem succeeds uniformly for every $\rho \geq 2 \log N$, including the boundary itself. First-order ML achievability was already known from the works just cited; the new point is that an explicit polynomial-time detector reaches the same scale.

Among convex relaxations, semidefinite and box relaxations provide the closest prior comparisons. For semidefinite relaxation, Kisialiou and Luo [2010]; So [2010] establish probabilistic approximation guarantees for the objective value, Jaldén and Ottersten [2008] show that the relaxation attains the maximum-likelihood diversity order in the fixed-dimensional high-SNR limit, and Lu et al. [2019]; Jiang et al. [2021] give explicit deterministic tightness conditions. None of these yields block-perfect recovery at any logarithmic SNR scale in the square real Gaussian model. For the box relaxation, Thrampoulidis et al. [2018] derive precise large-system symbol-error asymptotics at fixed SNR, and, extending the analysis to the vanishing-noise regime, Hu and Lu [2020] prove a Poisson limit for the number of errors and a sharp block-recovery transition: in the normalization used here, their square-system first-order threshold is $4 \log N$, with refined window $4 \log N - 2 \log \log N + O(1)$, a factor of two above maximum likelihood at first order. To our knowledge, this is the only previous theorem establishing polynomial-time block-perfect recovery at logarithmic SNR in this model.

The algorithm analyzed here is a steepest-selection variant of likelihood-ascent search (LAS). One-symbol LAS iteratively accepts likelihood-improving bit changes and has long been studied as a low-complexity approximation to ML detection [Sun, 1998; Vardhan et al., 2008]. Most directly, Mohammed et al. [2008] analyze one-symbol LAS in the square large-system limit, allowing MMSE initialization and using it in their simulations, and state asymptotic agreement with ML

error performance. That work does not establish the growing-SNR block-recovery result needed here: a uniform landscape guarantee over every error set that an adaptive trajectory may visit, together with confinement and a high-probability iteration bound.

Several rigorous algorithmic results are close in a different direction. The generalized power method of [Liu et al. \[2017\]](#) has a finite-iteration exact-recovery guarantee, but its random-channel theorem requires $m/n \geq 20/\beta^2$ with $\beta < 1/4$, hence a strongly overdetermined rather than square system. The zonotope algorithm of [Khan and Slock \[2003\]](#) computes exact ML in polynomial time when the receive rank is fixed; its enumeration is not polynomial when that rank grows as N .

Message-passing and statistical-physics analyses address an adjacent but different asymptotic question. The classical state-evolution theorem of [Bayati and Montanari \[2011\]](#) shows that, for each iteration index t held fixed as the dimensions tend to infinity, empirical averages of the AMP iterates converge to their scalar state-evolution predictions. MIMO specializations use state evolution to characterize large-system symbol-error rates at fixed SNR, system load, and constellation; under additional fixed-point conditions, they identify regimes in which LAMA attains the error rate of the individually optimal detector [[Jeon et al., 2018](#)]. Replica calculations predicted related Bayes-optimal large-system formulas for CDMA and Gaussian linear estimation [[Tanaka, 2002](#); [Guo and Verdú, 2005](#)]. Related replica-symmetric formulas for mutual information and Bayes-optimal estimation errors were subsequently established rigorously for broad Gaussian linear and generalized-linear models by [Reeves and Pfister \[2019\]](#) and [Barbier et al. \[2019\]](#). None of this implies block-perfect recovery: normalized error rates cannot distinguish zero errors from $o(N)$ errors, whereas block recovery requires exactly zero errors in the regime $\rho = \Theta(\log N)$.

The CLuP line of [Stojnic \[2019a,b\]](#) analyzes fixed-SNR bit-error performance and individual iterations through random-duality calculations and reports a small empirical iteration count. It does not establish exact block recovery at scaling SNR or a polynomial bound on the total number of iterations. Thus its broad title does not imply the four-part theorem proved here. Recent leave-one-out theory gives nonasymptotic coordinatewise representations for AMP [[Bao et al., 2025](#)] and may support a different proof route, but it does not state the present block theorem. Such a specialization would still have to handle the triangular growing-SNR regime, a possibly growing iteration count, leave-column variance calibration, and simultaneous Gaussian tails at the $1/N$ scale. We are not aware of a published specialization carrying out these steps at $\rho = 2 \log N$. MCMC analyses show that, at a suitably chosen temperature, the optimum receives inverse-polynomial (rather than exponentially small) stationary mass on the $2 \log N$ SNR scale, but polynomial mixing for square Gaussian instances remains open [[Hassibi et al., 2010, 2014](#)].

Our contributions. Our main result establishes that there is no first-order computational-statistical gap for exact recovery in the square Gaussian model. Rounded linear MMSE followed by a steepest-selection variant of one-bit likelihood-ascent search [[Sun, 1998](#); [Vardhan et al., 2008](#)] recovers the transmitted word throughout the regime $\rho \geq 2 \log N$, including the boundary $\rho = 2 \log N$ itself. More precisely, writing $\hat{x}_N$ for the detector output,

$$\lim_{N \rightarrow \infty} \sup_{\substack{x^* \in \{\pm 1\}^N \\ \rho \geq 2 \log N}} \Pr_{H,w} \{ \hat{x}_N \neq x^* \} = 0.$$

The detector accepts at most $O(N \log N)$ bit flips and uses $O(N^3)$ unit-cost exact-real arithmetic operations, with the dense LMMSE solve dominating the complexity.

The detector itself is classical. Closely related LMMSE-initialized one-symbol search procedures have been studied through asymptotic, symbol-error, and local-optimality analyses [Mohammed et al., 2008; Sun, 2009]. Its two-stage architecture—first obtain a partial estimate, then refine it through local corrections—is also familiar from exact-recovery problems with quadratic observations [see, e.g., the discussion in Hajek et al., 2016]. The contribution here is the threshold-scale uniform analysis of that architecture for the square Gaussian MIMO objective. Fixed-candidate and empirical symbol-error arguments do not control the data-dependent sequence of points visited by a local-search algorithm. We instead prove geometric bounds that hold simultaneously for every error set in a sufficiently large Hamming ball. These uniform bounds show that every nontruth point in the ball admits a decreasing bit flip and that the decreasing-cost trajectory cannot leave the ball. They therefore apply directly to the local search path, without conditioning on that path, and yield an explicit iteration bound at $\rho = 2 \log N$.

We complement the constructive result with an ML converse. Let $s_N \rightarrow \infty$ with $s_N = o(\log N)$. Uniformly over deterministic transmitted words and

$$0 < \rho \leq 2 \log N - \log \log N - s_N,$$

a one-bit neighbor has strictly smaller least-squares cost than the transmitted word with probability tending to one. Consequently, even ML detection fails in this regime. The achievability and converse boundaries differ by only $o(\log N)$, establishing the sharp first-order threshold $2 \log N$. A standard uniform-prior MAP argument, deferred to Appendix A, upgrades this ML converse to a minimax lower bound for arbitrary detectors.

Organization. The proof is presented in dependency order. We first establish that rounded LMMSE produces an estimate at sublinear Hamming distance from the transmitted word. We then derive the exact identities governing a one-bit flip and prove two uniform geometric estimates: a coarse estimate for the outer portion of the Hamming ball and a sharper estimate near the truth. Once these probabilistic events have been established, the remainder of the recovery argument is deterministic: the objective decreases, the trajectory cannot cross the boundary of the controlled ball, and every nontruth point offers an admissible decreasing move. We prove the ML converse after completing the constructive argument. Each lemma is introduced by the question it answers and by its role in the subsequent step. Appendix B collects exact non-MIMO formulations of the model and comparisons with the closest results in adjacent literatures.

2 System Setting and Main Result

Let

$$\mathbf{y} = \sqrt{\frac{\rho}{N}} \mathbf{H} \mathbf{x}^* + \mathbf{w}, \quad \mathbf{x}^* \in \{\pm 1\}^N, \quad (1)$$

where $\mathbf{H} \in \mathbb{R}^{N \times N}$ has independent $\mathcal{N}(0, 1)$ entries and $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N)$ is independent of $\mathbf{H}$. The transmitted vector $\mathbf{x}^*$ is deterministic and arbitrary. The receiver knows $(\mathbf{y}, \mathbf{H}, \rho)$. With this normalization, ρ is the average received SNR per observation.

Throughout, $N \in \{2, 3, \dots\}$ and $\rho > 0$; every asymptotic statement concerns $N \rightarrow \infty$. All logarithms are natural, and all probabilities refer only to the randomness of $(\mathbf{H}, \mathbf{w})$, unless explicitly stated otherwise. Complexity is measured in a unit-cost exact-real arithmetic model:

additions, subtractions, multiplications, divisions, comparisons, and evaluation of the known scale $\sqrt{\rho/N}$ are exact unit-cost operations. We do not claim a Turing bit-complexity or finite-precision bound.

The least-squares objective is

$$f(\mathbf{x}) = \left\| \mathbf{y} - \sqrt{\frac{\rho}{N}} \mathbf{H} \mathbf{x} \right\|_2^2, \quad \mathbf{x} \in \{\pm 1\}^N. \quad (2)$$

An ML detector returns any minimizer

$$\hat{\mathbf{x}}_{\text{ML}} \in \arg \min_{\mathbf{x} \in \{\pm 1\}^N} f(\mathbf{x}). \quad (3)$$

The polynomial-time detector analyzed in this paper has two stages.

Stage 1: rounded LMMSE. Compute

$$\mathbf{b} = \sqrt{\frac{N}{\rho}} \left(\mathbf{H}^\top \mathbf{H} + \frac{N}{\rho} \mathbf{I}_N \right)^{-1} \mathbf{H}^\top \mathbf{y}, \quad \mathbf{x}^{\text{init}} = \text{sign}(\mathbf{b}). \quad (4)$$

A zero coordinate is rounded to +1. Such a tie occurs with probability zero under the Gaussian model, so the convention does not affect the theorem. The positive prefactor $\sqrt{N/\rho}$ does not affect the rounded vector, but it is retained so that $\mathbf{b}$ is the correctly normalized LMMSE estimate.

Stage 2: steepest single-bit descent with a deterministic flip limit. For the current vector $\mathbf{x}$, let $\mathbf{x}^{(i)}$ denote the vector obtained by flipping coordinate i , and define its objective decrease by

$$\Delta_i(\mathbf{x}) = f(\mathbf{x}) - f(\mathbf{x}^{(i)}).$$

If

$$\max_{1 \leq i \leq N} \Delta_i(\mathbf{x}) < \frac{1}{N},$$

return the current vector. Otherwise, flip the smallest index attaining the largest gain. Define the deterministic flip limit

$$L_N = \min\{\ell \in \mathbb{N} : \ell \geq 1, 2^\ell \geq N\}, \quad M_N = 4NL_N.$$

Thus $L_N = \max\{1, \lceil \log_2 N \rceil\}$, and it can be computed by repeated integer doubling and comparison within the stated arithmetic model. After accepting the M_N -th flip, return the resulting vector unconditionally. Denote the vector returned in either case by $\hat{\mathbf{x}}$.

Once $\mathbf{H}^\top \mathbf{H}$ and the initial gains have been computed, all N gains can be updated in $O(N)$ operations after each accepted flip. Since the flip limit is $O(N \log N)$, the descent stage costs $O(N^2 \log N)$ operations. Forming and solving the dense LMMSE system costs $O(N^3)$ operations and therefore still dominates the total complexity.

Theorem 2.1 (Sharp first-order exact-recovery threshold). *The following two statements hold.*

(a) *The detector above obeys the uniform error-probability guarantee*

$$\lim_{N \rightarrow \infty} \sup_{\substack{\mathbf{x}^* \in \{\pm 1\}^N \\ \rho \geq 2 \log N}} \mathbb{P}_{H,w} \{\hat{\mathbf{x}} \neq \mathbf{x}^*\} = 0. \quad (5)$$

Its worst-case cost is $O(N^3)$ unit-cost exact-real arithmetic operations.

(b) *Let $s_N \rightarrow \infty$ with $s_N = o(\log N)$. Then*

$$\lim_{N \rightarrow \infty} \sup_{\substack{\mathbf{x}^* \in \{\pm 1\}^N \\ 0 < \rho \leq 2 \log N - \log \log N - s_N}} \mathbb{P}_{H,w} \{\mathbf{x}^* \text{ is an ML minimizer of } f\} = 0. \quad (6)$$

Consequently, the block-error probability of every ML detector tends to one uniformly throughout this lower regime.

For all sufficiently large N , the upper endpoint in (6) is positive, so its parameter set is nonempty.

The supremum in (5) concerns the worst error probability over the displayed deterministic parameters. It does not assert that one realization of (H, w) succeeds simultaneously for every transmitted word and every SNR. Likewise, (6) is a uniform probability statement, not a simultaneous assertion over all words and SNR values on one realization.

The gap between the two proven boundaries is $\log \log N + s_N = o(\log N)$. The theorem therefore identifies the common first-order constant 2, but does not claim to determine the complete lower-order transition window. The decision-theoretic minimax interpretation is recorded separately in Appendix A.

3 Roadmap of the achievability proof

The proof is easiest to read as a sequence of questions. How close does rounded LMMSE get to the truth? What does the objective landscape look like inside that neighborhood? Once the landscape bounds hold, why can the greedy path neither escape, stop incorrectly, nor exhaust its allowed number of flips before recovery? The three panels below answer these questions in the same order as the formal proof.

A. Rounded LMMSE comes within sublinear distance of the ground truth

1. Convert estimation error into one trace. With $B_\rho = (I_N + \frac{\rho}{N} H^\top H)^{-1}$, the exact LMMSE identity is

$$\mathbb{E} \|\mathbf{b} - \mathbf{x}^*\|_2^2 = \mathbb{E} \operatorname{tr}(B_\rho).$$

Lemma 5.1.

2. Bound that trace. A leave-one-row identity and a direct expectation bound give, uniformly for $\rho \geq 2 \log N$,

$$\mathbb{E} \operatorname{tr}(B_\rho) \leq \frac{2N}{\sqrt{\log N}}.$$

Lemmas 5.2 and 5.3.

3. Convert squared error into sign errors. Define

$$\mathbf{x}^{\text{init}} = \operatorname{sign}(\mathbf{b}), \quad K = d_H(\mathbf{x}^{\text{init}}, \mathbf{x}^*).$$

If coordinate j is rounded incorrectly, then $|b_j - x_j^*| \geq 1$. Therefore every wrong coordinate contributes at least one to the squared error, and hence $K \leq \|\mathbf{b} - \mathbf{x}^*\|_2^2$. Consequently, Markov's inequality and the preceding trace bound give

$$\begin{aligned} \Pr \left\{ K > \frac{N}{(\log N)^{1/4}} \right\} &\leq \Pr \left\{ \|\mathbf{b} - \mathbf{x}^*\|_2^2 > \frac{N}{(\log N)^{1/4}} \right\} \\ &\leq \frac{\mathbb{E} \|\mathbf{b} - \mathbf{x}^*\|_2^2}{N / (\log N)^{1/4}} \leq \frac{2}{(\log N)^{1/4}} \rightarrow 0. \end{aligned}$$

Proposition 5.4.

B. The landscape near the ground truth has favorable geometry

For a current vector x , set $S = \{i : x_i \neq x_i^*\}$, $k = |S|$, and $h_i = x_i^* H_{:,i}$, using column-sign symmetry to normalize $x^* = \mathbf{1}$. This sign is needed in the noise terms. Ceilings are suppressed only in this roadmap.

Exact flip algebra: the two quantities we need

$$f(x) - f(x^*) = \frac{4\rho}{N} \left\| \sum_{i \in S} h_i \right\|_2^2 + 4\sqrt{\frac{\rho}{N}} w^\top \sum_{i \in S} h_i,$$

$$\sum_{i \in S} \Delta_i(x) = 4 \left[\frac{\rho}{N} \left(2 \left\| \sum_{i \in S} h_i \right\|_2^2 - \sum_{i \in S} \|h_i\|_2^2 \right) + \sqrt{\frac{\rho}{N}} w^\top \sum_{i \in S} h_i \right].$$

The first line is the excess cost. The second sums the gains from correcting all k wrong bits: if it is at least ak , some wrong bit has gain at least a , so the maximal available gain is no smaller; if $a \geq 1/N$, the detector accepts such a gain. [Lemmas 6.1 to 6.3](#).

Outer region: coarse bounds suffice

$$\frac{N}{(\log N)^3} < k \leq \frac{4N}{(\log N)^{1/4}}.$$

With probability tending to one, simultaneously for every such set,

$$\frac{5}{2}\rho k \leq f(x) - f(x^*) \leq \frac{11}{2}\rho k,$$

$$\max_{i \in S} \Delta_i(x) \geq \frac{\rho}{2}.$$

Uniformity mechanism. Conditional on the uniform channel bounds, fixed-set failure is at most

$$2e^{-k \log N / 80}.$$

There are exactly $\binom{N}{k}$ sets and

$$\log \binom{N}{k} \leq k \log \frac{eN}{k} \leq k(1 + 3 \log \log N).$$

This exponent is $o(k \log N)$, so the union bound tends to zero. [Lemmas 7.1 to 7.3](#).

Inner region: sharp tails are required

$$1 \leq k \leq \frac{N}{(\log N)^3}.$$

With probability tending to one, simultaneously for every such set,

$$\frac{12\rho k}{(\log N)^{5/4}} \leq f(x) - f(x^*) \leq 8\rho k,$$

$$\max_{i \in S} \Delta_i(x) \geq \frac{4\rho}{(\log N)^{5/4}}.$$

Uniformity mechanism. Conditional on the sharper channel bounds, fixed-set failure is at most

$$\frac{2}{\sqrt{k \log N}} N^{-k(1-9/(\log N)^{5/4})}.$$

Since $\binom{N}{k} \leq N^k/k!$, the leading powers cancel; the residual correction is at most 2^k , and the full union bound is at most

$$\frac{2}{\sqrt{\log N}} \sum_{k=1}^{\lfloor N/(\log N)^3 \rfloor} \frac{2^k}{k! \sqrt{k}} \leq \frac{2(e^2 - 1)}{\sqrt{\log N}} \rightarrow 0.$$

[Lemmas 8.1 to 8.3](#).

What the two regions give together

The ranges meet at $N/(\log N)^3$, so every nontruth vector with $1 \leq d_H(x, x^*) \leq 4N/(\log N)^{1/4}$ is covered uniformly before the algorithm runs. Consequently,

$$\text{decreasing move: } \exists i \in S : \Delta_i(x) \geq \frac{4\rho}{(\log N)^{5/4}},$$

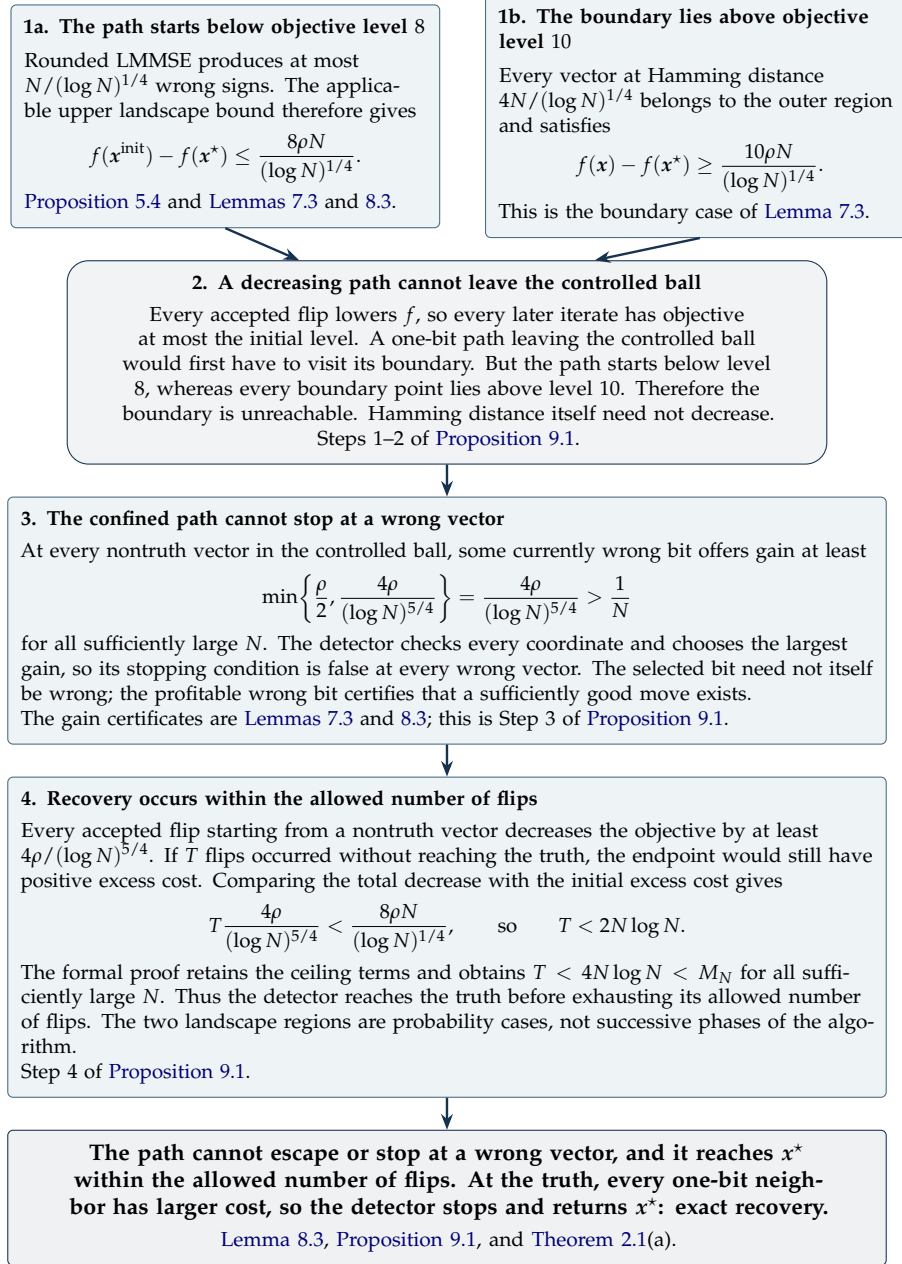
$$\text{outer barrier: } d_H(x, x^*) = \frac{4N}{(\log N)^{1/4}} \Rightarrow f(x) - f(x^*) \geq \frac{10\rho N}{(\log N)^{1/4}}.$$

Thus there is no wrong stopping point in the ball; a decreasing path that starts below $8\rho N/(\log N)^{1/4}$ cannot cross its boundary. The statements are uniform, so they apply along the adaptive path. [Lemmas 7.3 and 8.3; Proposition 9.1](#).

Warm start $\cap$ landscape. $\Pr\{\text{both hold}\} \geq 1 - \Pr\{\text{warm start fails}\} - \Pr\{\text{landscape fails}\} \rightarrow 1$. No independence is required.

C. A decreasing objective forces exact recovery

Fix a realization on which the warm-start and landscape events both hold. The remainder of the argument is deterministic.



4 Standard inequalities used in the proof

Throughout the paper, an event holds “with probability tending to one” if its probability, under the joint randomness of $\mathbf{H}$ and $\mathbf{w}$, converges to one as $N \rightarrow \infty$. Equivalently, the probability of its

complement tends to zero. Whenever useful below, we keep the explicit finite- N upper bound on that complementary probability rather than using only the asymptotic phrase.

We state the external ingredients before using them. This makes every later application identifiable by name.

S1. Markov. If $X \geq 0$ and $t > 0$, then $\mathbb{P}\{X \geq t\} \leq \mathbb{E}X/t$.

S2. McDiarmid's inequality. Suppose F is a function of independent inputs and replacing input i changes F by at most c_i . Then, for $t > 0$,

$$\mathbb{P}\{F - \mathbb{E}F \leq -t\} \leq \exp\left(-\frac{2t^2}{\sum_i c_i^2}\right). \quad (7)$$

S3. Union bound. For arbitrary events $A_1, \dots, A_m$, $\mathbb{P}(\cup_j A_j) \leq \sum_j \mathbb{P}(A_j)$.

S4. Subset count. For $1 \leq k \leq N$,

$$\binom{N}{k} \leq \frac{N^k}{k!} \leq \left(\frac{eN}{k}\right)^k. \quad (8)$$

S5. Gaussian and chi-square tails. If $Z \sim \mathcal{N}(0, 1)$, then

$$\mathbb{P}\{|Z| \geq t\} = 2Q(t) \leq \min\left\{2e^{-t^2/2}, \frac{2\phi(t)}{t}\right\}, \quad t > 0, \quad (9)$$

where $\phi(t) = (2\pi)^{-1/2}e^{-t^2/2}$ and $Q(t) = \int_t^\infty \phi(u) du$. If $X \sim \chi_N^2$ and $0 < \varepsilon < 1$, then

$$\mathbb{P}\{|X - N| > \varepsilon N\} \leq 2 \exp\left(-\frac{N\varepsilon^2}{8}\right). \quad (10)$$

S6. Gaussian quadratic-form lower tail. If $\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_N)$ and $\mathbf{0} \preceq \mathbf{M} \preceq \mathbf{I}_N$, then

$$\mathbb{P}\left\{\mathbf{g}^\top \mathbf{M} \mathbf{g} < \frac{1}{2} \operatorname{tr} \mathbf{M}\right\} \leq \exp\left(-\frac{\operatorname{tr} \mathbf{M}}{16}\right). \quad (11)$$

Indeed, the Laurent–Massart bound gives $\mathbf{g}^\top \mathbf{M} \mathbf{g} \geq \operatorname{tr} \mathbf{M} - 2\sqrt{\operatorname{tr}(\mathbf{M}^2)t}$ except with probability e^{-t} . Take $t = \operatorname{tr}(\mathbf{M})/16$ and use $\operatorname{tr}(\mathbf{M}^2) \leq \operatorname{tr} \mathbf{M}$.

S7. Rank-one inverse formula. For invertible $\mathbf{A}$ satisfying $1 + \mathbf{u}^\top \mathbf{A}^{-1} \mathbf{u} \neq 0$,

$$(\mathbf{A} + \mathbf{u}\mathbf{u}^\top)^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{u}^\top\mathbf{A}^{-1}}{1 + \mathbf{u}^\top\mathbf{A}^{-1}\mathbf{u}}. \quad (12)$$

Markov, McDiarmid's inequality, the union bound, and the subset estimate are standard; see, for example, [Boucheron et al. \[2013\]](#). The bound $Q(t) \leq \phi(t)/t$ in (9) is the classical Mills-ratio inequality [[Gordon, 1941](#)]; the exponential bound follows from the Gaussian Chernoff bound. The chi-square estimate follows from the chi-square moment-generating function; see [Laurent and Massart \[2000\]](#); the same reference gives the weighted quadratic-form lower tail used in (11). The rank-one identity is due to [Sherman and Morrison \[1950\]](#).

5 How close rounded LMMSE gets

The descent proof will need exactly one fact from its initialization:

$$d_H(\mathbf{x}^{\text{init}}, \mathbf{x}^*) \leq \left\lceil \frac{N}{(\log N)^{1/4}} \right\rceil \quad \text{with probability tending to one.} \quad (13)$$

We derive backward from this goal before proving anything about the random matrix.

The vector $\mathbf{b}$ in (4) is already normalized to estimate $\mathbf{x}^*$ directly. We therefore analyze $\mathbf{b}$ itself.

5.1 What is sufficient for the desired Hamming radius?

If coordinate j has the wrong sign, then $|b_j - x_j^*| \geq 1$. Therefore, for every realization,

$$d_H(\mathbf{x}^{\text{init}}, \mathbf{x}^*) \leq \|\mathbf{b} - \mathbf{x}^*\|_2^2. \quad (14)$$

Markov's inequality now shows what remains to be proved. Any estimate of the form

$$\mathbb{E} \|\mathbf{b} - \mathbf{x}^*\|_2^2 \leq C \frac{N}{\sqrt{\log N}} \quad (15)$$

with an absolute constant C implies

$$\mathbb{P} \left\{ d_H(\mathbf{x}^{\text{init}}, \mathbf{x}^*) > \frac{N}{(\log N)^{1/4}} \right\} \leq \frac{C}{(\log N)^{1/4}} \longrightarrow 0.$$

Thus the LMMSE stage does not require coordinatewise accuracy. We need only the total expected squared error at the scale $N / \sqrt{\log N}$.

Define the LMMSE error matrix

$$\mathbf{B}_\rho = \left(\mathbf{I}_N + \frac{\rho}{N} \mathbf{H}^\top \mathbf{H} \right)^{-1}. \quad (16)$$

5.2 The exact mean-square identity reduces the problem to a trace

Lemma 5.1 (Exact LMMSE mean-square error). *For every fixed $\mathbf{x}^* \in \{\pm 1\}^N$,*

$$\mathbb{E} \|\mathbf{b} - \mathbf{x}^*\|_2^2 = \mathbb{E} \text{tr}(\mathbf{B}_\rho). \quad (17)$$

Proof. From (16),

$$\left(\mathbf{H}^\top \mathbf{H} + \frac{N}{\rho} \mathbf{I}_N \right)^{-1} = \frac{\rho}{N} \mathbf{B}_\rho, \quad \frac{\rho}{N} \mathbf{B}_\rho \mathbf{H}^\top \mathbf{H} = \mathbf{I}_N - \mathbf{B}_\rho.$$

Substituting the model into the already normalized estimator (4) therefore gives

$$\mathbf{b} = (\mathbf{I}_N - \mathbf{B}_\rho) \mathbf{x}^* + \sqrt{\frac{\rho}{N}} \mathbf{B}_\rho \mathbf{H}^\top \mathbf{w}.$$

After subtracting $\mathbf{x}^*$,

$$\mathbf{b} - \mathbf{x}^* = -\mathbf{B}_\rho \mathbf{x}^* + \sqrt{\frac{\rho}{N}} \mathbf{B}_\rho \mathbf{H}^\top \mathbf{w}. \quad (18)$$

Condition on $\mathbf{H}$ and expand the squared norm. The cross term has conditional expectation zero because $\mathbb{E}(\mathbf{w}) = \mathbf{0}$. The noise term has conditional covariance

$$\begin{aligned} & \frac{\rho}{N} \mathbf{B}_\rho \mathbf{H}^\top \mathbf{H} \mathbf{B}_\rho \\ &= \mathbf{B}_\rho \left(\mathbf{B}_\rho^{-1} - \mathbf{I}_N \right) \mathbf{B}_\rho = \mathbf{B}_\rho - \mathbf{B}_\rho^2. \end{aligned}$$

Therefore

$$\mathbb{E}_w \left[\|\mathbf{b} - \mathbf{x}^*\|_2^2 \middle| \mathbf{H} \right] = \mathbf{x}^{*\top} \mathbf{B}_\rho^2 \mathbf{x}^* + \text{tr}(\mathbf{B}_\rho - \mathbf{B}_\rho^2). \quad (19)$$

It remains to average the first term over $\mathbf{H}$. For every fixed orthogonal matrix $\mathbf{O}$, the matrices $\mathbf{H}\mathbf{O}$ and $\mathbf{H}$ have the same distribution. Consequently, $\mathbb{E}(\mathbf{B}_\rho^2)$ commutes with every orthogonal matrix and must be a scalar multiple of $\mathbf{I}_N$. Its trace determines that scalar:

$$\mathbb{E}(\mathbf{B}_\rho^2) = \frac{\mathbb{E} \text{tr}(\mathbf{B}_\rho^2)}{N} \mathbf{I}_N.$$

Since $\|\mathbf{x}^*\|_2^2 = N$,

$$\mathbb{E} \left[\mathbf{x}^{*\top} \mathbf{B}_\rho^2 \mathbf{x}^* \right] = \mathbb{E} \text{tr}(\mathbf{B}_\rho^2).$$

Average (19) over $\mathbf{H}$. The positive and negative copies of $\mathbb{E} \text{tr}(\mathbf{B}_\rho^2)$ cancel, leaving (17). $\square$

Reduction to a trace bound. Equations (14) and (17) reduce the entire warm-start claim to one question:

$$\text{Can we prove} \quad \mathbb{E} \text{tr}(\mathbf{B}_\rho) = O \left(\frac{N}{\sqrt{\log N}} \right)?$$

No coordinatewise LMMSE analysis is needed.

5.3 From a leave-one-row identity to a trace bound

Increasing ρ increases the positive matrix inside the inverse in (16), and therefore decreases its inverse. Thus, for $\rho \geq 2 \log N$,

$$\mathbf{B}_\rho \preceq \left(\mathbf{I}_N + \frac{2 \log N}{N} \mathbf{H}^\top \mathbf{H} \right)^{-1}.$$

For this subsection only, write

$$\mathbf{B} = \left(\mathbf{I}_N + \frac{2 \log N}{N} \mathbf{H}^\top \mathbf{H} \right)^{-1}. \quad (20)$$

It is enough to bound $\mathbb{E} \text{tr}(\mathbf{B})$.

A direct row-by-row calculation with $\mathbf{B}$ would be invalid: row $\mathbf{H}_{i,:}$ already appears inside $\mathbf{B}$. We therefore remove that row first and define

$$\mathbf{B}_{-i} = \left(\mathbf{I}_N + \frac{2 \log N}{N} \sum_{\ell \neq i} \mathbf{H}_{\ell,:}^\top \mathbf{H}_{\ell,:} \right)^{-1}, \quad (21)$$

$$q_i = \frac{2 \log N}{N} \mathbf{H}_{i,:} \mathbf{B}_{-i} \mathbf{H}_{i,:}^\top. \quad (22)$$

Now $\mathbf{B}_{-i}$ depends only on the other rows and is independent of $\mathbf{H}_{i,:}$. The scalar q_i is not computed by the detector. It is the quadratic form that appears in the Sherman–Morrison denominator when row i is restored. Conditional on $\mathbf{B}_{-i}$, its unscaled quadratic part has mean $\text{tr}(\mathbf{B}_{-i})$; this independence is what will connect the random denominator to the trace we want to bound.

Lemma 5.2 (The trace identity). *For every realization,*

$$\text{tr}(\mathbf{B}) = \sum_{i=1}^N \frac{1}{1 + q_i}, \quad 0 \leq \text{tr}(\mathbf{B}_{-i}) - \text{tr}(\mathbf{B}) \leq 1. \quad (23)$$

More generally, when one rank-one positive-semidefinite term is added to or removed from a matrix of the form $\mathbf{I}_N + \sum_j \mathbf{u}_j \mathbf{u}_j^\top$, the traces of the corresponding inverses differ by at most one, regardless of the number of terms in the sum.

Proof. Adding row i adds the rank-one matrix $(2 \log N / N) \mathbf{H}_{i,:}^\top \mathbf{H}_{i,:}$ to $\mathbf{B}_{-i}^{-1}$. Apply the rank-one inverse formula (12). It gives

$$\frac{2 \log N}{N} \mathbf{H}_{i,:} \mathbf{B} \mathbf{H}_{i,:}^\top = \frac{q_i}{1 + q_i}.$$

Now use $\mathbf{B}^{-1} \mathbf{B} = \mathbf{I}_N$, take the trace, and substitute the preceding equality:

$$\begin{aligned} N &= \text{tr}(\mathbf{B}) + \sum_{i=1}^N \frac{2 \log N}{N} \mathbf{H}_{i,:} \mathbf{B} \mathbf{H}_{i,:}^\top \\ &= \text{tr}(\mathbf{B}) + \sum_{i=1}^N \frac{q_i}{1 + q_i} \\ &= \text{tr}(\mathbf{B}) + N - \sum_{i=1}^N \frac{1}{1 + q_i}. \end{aligned}$$

Cancel N to obtain the first identity.

The same rank-one formula gives

$$\text{tr}(\mathbf{B}_{-i}) - \text{tr}(\mathbf{B}) = \frac{(2 \log N / N) \mathbf{H}_{i,:} \mathbf{B}_{-i}^2 \mathbf{H}_{i,:}^\top}{1 + q_i}.$$

Because $\mathbf{0} \preceq \mathbf{B}_{-i} \preceq \mathbf{I}_N$, we have $\mathbf{B}_{-i}^2 \preceq \mathbf{B}_{-i}$. The numerator is therefore between zero and q_i , proving that the ratio lies in $[0, 1]$. This last calculation uses only that the inverse before the rank-one update is bounded above by $\mathbf{I}_N$. It therefore proves the stated general trace-sensitivity bound for any number of pre-existing rank-one terms. $\square$

The trace identity also suggests how to prove the correct scale. Conditional on $\mathbf{B}_{-i}$, Gaussian quadratic-form concentration predicts

$$\mathbf{H}_{i,:} \mathbf{B}_{-i} \mathbf{H}_{i,:}^\top \approx \text{tr}(\mathbf{B}_{-i}),$$

while removing one row changes the trace by at most one. Thus one expects

$$q_i \approx \frac{2 \log N}{N} \text{tr}(\mathbf{B}).$$

If this approximation is inserted into the exact identity (23), it yields the self-consistent relation

$$\text{tr}(\mathbf{B}) \approx \frac{N}{1 + (2 \log N / N) \text{tr}(\mathbf{B})},$$

whose positive solution has order $N / \sqrt{\log N}$. This calculation is not used as a proof; it identifies the estimate to establish. The next lemma replaces the two approximations by lower-tail bounds and derives a rigorous inequality for $\mathbb{E} \text{tr}(\mathbf{B})$.

Lemma 5.3 (Direct expected-trace bound). *For all sufficiently large N ,*

$$\mathbb{E} \text{tr}(\mathbf{B}) \leq \frac{2N}{\sqrt{\log N}}. \quad (24)$$

Proof. Write $m = \mathbb{E} \text{tr}(\mathbf{B})$. We turn the heuristic preceding the lemma into a scalar inequality for m .

Step 1: reduce the trace identity to one representative row. The random variables $q_1, \dots, q_N$ are identically distributed because the rows of $\mathbf{H}$ are exchangeable. Taking expectations in (23) therefore gives

$$\frac{m}{N} = \mathbb{E} \left[\frac{1}{1 + q_1} \right]. \quad (25)$$

Thus it is enough to show that q_1 is usually large. The leave-one-row construction makes this possible because $\mathbf{B}_{-1}$ is independent of the Gaussian row appearing in q_1 .

Step 2: the inverse built from the other rows usually has a large trace. Deleting a positive-semidefinite rank-one term can only increase the inverse in the positive-semidefinite order. Consequently,

$$\mathbb{E} \text{tr}(\mathbf{B}_{-1}) \geq m.$$

Replacing any one of the remaining rows changes $\text{tr}(\mathbf{B}_{-1})$ by at most two: delete the old row, which changes the trace by at most one by (23), and add the new row, which changes it by at most one again. Hence the bounded-difference constants are at most two. Applying McDiarmid's inequality at a downward deviation of $m/2$, and using $\mathbb{E} \text{tr}(\mathbf{B}_{-1}) \geq m$, gives

$$\mathbb{P} \left\{ \text{tr}(\mathbf{B}_{-1}) < \frac{m}{2} \right\} \leq \exp \left(-\frac{m^2}{8N} \right). \quad (26)$$

Let

$$A = \left\{ \text{tr}(\mathbf{B}_{-1}) \geq \frac{m}{2} \right\}.$$

Step 3: the independent Gaussian row produces a large quadratic form. Condition on $\mathbf{B}_{-1}$. This matrix is now fixed, whereas $\mathbf{H}_{1,:}$ remains a standard Gaussian row. Substituting $\mathbf{g} = \mathbf{H}_{1,:}^\top$ and $\mathbf{M} = \mathbf{B}_{-1}$ into (11) is valid because $\mathbf{0} \preceq \mathbf{B}_{-1} \preceq \mathbf{I}_N$. Hence

$$\begin{aligned} \mathbb{P} \left\{ \mathbf{H}_{1,:} \mathbf{B}_{-1} \mathbf{H}_{1,:}^\top < \frac{1}{2} \text{tr}(\mathbf{B}_{-1}) \mid \mathbf{B}_{-1} \right\} \\ \leq \exp \left(-\frac{\text{tr}(\mathbf{B}_{-1})}{16} \right) \leq \exp \left(-\frac{m}{32} \right) \end{aligned} \quad (27)$$

on the event A , where the last inequality uses $\text{tr}(\mathbf{B}_{-1}) \geq m/2$. Let

$$C = \left\{ \mathbf{H}_{1,:} \mathbf{B}_{-1} \mathbf{H}_{1,:}^\top \geq \frac{1}{2} \text{tr}(\mathbf{B}_{-1}) \right\}.$$

Equation (27) says that, conditional on any $\mathbf{B}_{-1}$ for which A holds, C fails with probability at most $e^{-m/32}$.

Step 4: convert the two lower bounds into a reciprocal bound. Call $G = A \cap C$ the good event. On G , first use C , then A , in the definition of q_1 :

$$\begin{aligned} q_1 &= \frac{2 \log N}{N} \mathbf{H}_{1,:} \mathbf{B}_{-1} \mathbf{H}_{1,:}^\top \\ &\geq \frac{2 \log N}{N} \cdot \frac{1}{2} \text{tr}(\mathbf{B}_{-1}) \geq \frac{2 \log N}{4N} m. \end{aligned} \quad (28)$$

The function $u \mapsto (1+u)^{-1}$ is decreasing on $[0, \infty)$. Therefore (28) gives, on G ,

$$\frac{1}{1+q_1} \leq \frac{1}{1 + (2 \log N / 4N) m}.$$

This is the reciprocal inequality used below: it follows directly from the preceding lower bound on q_1 .

Step 5: control the complement of the good event. The event G requires both A and C . It therefore fails in exactly one of the following disjoint ways:

$$\begin{aligned} \text{(i)} \quad A^c : \quad \text{tr}(\mathbf{B}_{-1}) < \frac{m}{2}, \\ \text{(ii)} \quad A \cap C^c : \quad \text{tr}(\mathbf{B}_{-1}) \geq \frac{m}{2} \quad \text{but} \quad \mathbf{H}_{1,:} \mathbf{B}_{-1} \mathbf{H}_{1,:}^\top < \frac{1}{2} \text{tr}(\mathbf{B}_{-1}). \end{aligned}$$

The first probability is bounded by (26). Averaging the conditional estimate (27) over deleted matrices in A bounds the second. Thus

$$\mathbb{P}(G^c) \leq e^{-m^2/(8N)} + e^{-m/32}.$$

Also $q_1 \geq 0$, because $\mathbf{B}_{-1} \succeq \mathbf{0}$, so $(1+q_1)^{-1} \leq 1$ on G^c . Splitting the expectation in (25) over G and G^c now yields

$$\begin{aligned} \frac{m}{N} &\leq \frac{1}{1 + (2 \log N / 4N) m} \\ &\quad + \exp \left(-\frac{m^2}{8N} \right) + \exp \left(-\frac{m}{32} \right). \end{aligned} \quad (29)$$

Step 6: solve the scalar inequality. We now extract $m \leq 2N / \sqrt{\log N}$ from (29). The comparison value $4\sqrt{N \log N}$ separates two easy cases.

If

$$m \leq 4\sqrt{N \log N},$$

then the desired bound already holds for all sufficiently large N , because

$$4\sqrt{N \log N} \leq \frac{2N}{\sqrt{\log N}} \iff 2 \log N \leq \sqrt{N}.$$

It remains to consider

$$m > 4\sqrt{N \log N}.$$

In this case,

$$\exp\left(-\frac{m^2}{8N}\right) \leq e^{-2 \log N} = \frac{1}{N^2}.$$

Also, for sufficiently large N ,

$$\exp\left(-\frac{m}{32}\right) \leq \exp\left(-\frac{\sqrt{N \log N}}{8}\right) \leq \frac{1}{N^2}.$$

Thus the two failure contributions together are at most $2/N^2$. The remaining reciprocal term is bounded using $1/(1+u) \leq 1/u$. Multiplying the resulting inequality by N gives

$$m \leq \frac{4N^2}{(2 \log N)m} + \frac{2}{N}.$$

Multiply by m and use $m \leq N$, which follows from $\mathbf{0} \preceq \mathbf{B} \preceq \mathbf{I}_N$:

$$m^2 \leq \frac{4N^2}{2 \log N} + 2.$$

This is again at most $4N^2 / \log N$ for sufficiently large N , which is (24) after taking square roots. $\square$

The trace estimate now has exactly the scale required in (15). It remains to combine it with the exact MSE identity and the deterministic implication from a wrong sign to unit squared error.

5.4 Convert the trace bound into a Hamming-distance warm start

Proposition 5.4 (Rounded LMMSE enters the outer region). *For all sufficiently large N , uniformly over $\mathbf{x}^* \in \{\pm 1\}^N$ and $\rho \geq 2 \log N$,*

$$\mathbb{P} \left\{ d_H(\mathbf{x}^{\text{init}}, \mathbf{x}^*) > \left\lceil \frac{N}{(\log N)^{1/4}} \right\rceil \right\} \leq \frac{2}{(\log N)^{1/4}} \longrightarrow 0. \quad (30)$$

Proof. The exact LMMSE calculation proved

$$\mathbb{E} \|\mathbf{b} - \mathbf{x}^*\|_2^2 = \mathbb{E} \text{tr}(\mathbf{B}_\rho).$$

For every $\rho \geq 2 \log N$, increasing the positive matrix inside the inverse gives

$$\mathbf{B}_\rho \preceq \mathbf{B} = \left(\mathbf{I}_N + \frac{2 \log N}{N} \mathbf{H}^\top \mathbf{H} \right)^{-1}.$$

The trace lemma just proved $\mathbb{E} \operatorname{tr}(\mathbf{B}) \leq 2N / \sqrt{\log N}$. Combining these three statements yields

$$\mathbb{E} \|\mathbf{b} - \mathbf{x}^\star\|_2^2 \leq \frac{2N}{\sqrt{\log N}}.$$

We also proved the following pointwise implication: every wrong rounded sign contributes at least one to the squared error. Hence, for every realization,

$$d_H(\mathbf{x}^{\text{init}}, \mathbf{x}^\star) \leq \|\mathbf{b} - \mathbf{x}^\star\|_2^2.$$

Markov's inequality at the threshold $N / (\log N)^{1/4}$ now gives

$$\begin{aligned} \mathbb{P} \left\{ d_H(\mathbf{x}^{\text{init}}, \mathbf{x}^\star) > \frac{N}{(\log N)^{1/4}} \right\} &\leq \mathbb{P} \left\{ \|\mathbf{b} - \mathbf{x}^\star\|_2^2 > \frac{N}{(\log N)^{1/4}} \right\} \\ &\leq \frac{2N / \sqrt{\log N}}{N / (\log N)^{1/4}} = \frac{2}{(\log N)^{1/4}}. \end{aligned}$$

The ceiling can only enlarge the allowed set.

This probability tends to zero slowly, but that causes no later loss. In the final theorem it is added to the failure probabilities of the uniform landscape events. Once all those events hold, the descent proof is deterministic; the warm-start probability is never multiplied by the number of iterations or used in a denominator. $\square$

Stage 1 is complete. Rounded LMMSE need not recover every bit. Its only job is to enter the ball of radius $N / (\log N)^{1/4}$, and the exact MSE identity plus Markov does this directly. The exponent $1/4$ is deliberately loose: the same argument gives a high-probability radius $(N / \sqrt{\log N}) g_N$ for any $g_N \rightarrow \infty$, but a tighter radius does not simplify or strengthen the final exact-recovery theorem.

6 Exact algebra for objective descent

Rounded LMMSE has placed us inside a controlled Hamming ball. Our remaining goal is to show that greedy descent reaches the truth in $O(N \log N)$ accepted flips, before exhausting the detector's allowed number of flips.

One cannot prove this by claiming that every accepted flip corrects a wrong bit. The detector chooses the largest objective decrease among all coordinates, and that coordinate need not always be one of the currently wrong coordinates. Hamming distance may therefore increase at an intermediate step. The quantity that decreases monotonically is the least-squares objective f .

The descent proof consequently needs two facts at every nontruth vector in the controlled ball:

1. The current objective is above the objective at the truth. Upper and lower bounds on this objective gap will confine the path and give a total budget for the number of flips.
2. At least one wrong bit offers a definite objective decrease. The greedy detector chooses the best flip among all bits, so its chosen decrease is at least as large.

This section derives the exact identities from which [Sections 7](#) and [8](#) will prove these two facts.

6.1 Normalize the truth and encode the current errors

The channel distribution is invariant under column sign changes. We first use this symmetry to normalize the transmitted word, and then represent any current iterate solely by the set of coordinates on which it is wrong. This notation makes the two quantities needed by descent explicit.

Multiplying column i of $\mathbf{H}$ by x_i^* preserves its standard Gaussian distribution. It maps the transmitted vector to $\mathbf{1}$, preserves Hamming distance, and maps every bit-flip gain to the corresponding gain in the transformed problem. The LMMSE initialization transforms in the same way almost surely. Indeed, the unrounded LMMSE coordinates have continuous distributions, so the convention used for $\text{sign}(0)$ is encountered with probability zero. We may therefore prove the remaining statements for

$$\mathbf{x}^* = \mathbf{1}.$$

For a current vector $\mathbf{x}$, let

$$S = \{i : x_i = -1\} \tag{31}$$

be its set of wrong coordinates, and write $\mathbf{h}_i$ for column i of $\mathbf{H}$. Every coordinate in S equals -1 instead of $+1$, so its error has size two. Therefore

$$\mathbf{H}(\mathbf{1} - \mathbf{x}) = 2 \sum_{i \in S} \mathbf{h}_i.$$

Substituting the channel model gives the current residual:

$$\begin{aligned} \mathbf{y} - \sqrt{\frac{\rho}{N}} \mathbf{H} \mathbf{x} &= \mathbf{w} + \sqrt{\frac{\rho}{N}} \mathbf{H}(\mathbf{1} - \mathbf{x}) \\ &= \mathbf{w} + 2 \sqrt{\frac{\rho}{N}} \sum_{i \in S} \mathbf{h}_i. \end{aligned} \tag{32}$$

It is the original noise plus the channel contribution of the wrong bits.

6.2 First question: how far above the truth is the objective?

At the truth, the residual is $\mathbf{w}$, and hence

$$f(\mathbf{1}) = \|\mathbf{w}\|_2^2.$$

The following identity compares the current objective with that value.

Lemma 6.1 (Exact objective gap). *For every error set S ,*

$$f(\mathbf{x}) - f(\mathbf{1}) = \frac{4\rho}{N} \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 + 4\sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i. \quad (33)$$

Proof. Insert the residual from (32), expand the square, and subtract the objective at the truth:

$$\begin{aligned} f(\mathbf{x}) - f(\mathbf{1}) &= \left\| \mathbf{w} + 2\sqrt{\frac{\rho}{N}} \sum_{i \in S} \mathbf{h}_i \right\|_2^2 - \|\mathbf{w}\|_2^2 \\ &= \frac{4\rho}{N} \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 + 4\sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i. \end{aligned}$$

□

The first term is the positive signal penalty caused by the wrong bits. For a typical Gaussian channel it is approximately $4\rho|S|$. The second term is the projection of the noise onto the same sum of channel columns; it can have either sign. The later probability lemmas will show uniformly that the noise term cannot cancel the signal penalty. That will place every controlled nontruth vector strictly above the truth.

6.3 Second question: does some wrong bit offer progress?

For $i \in S$, let $\mathbf{x}^{(i)}$ be the vector obtained by correcting wrong bit i . Correcting it subtracts $2\sqrt{\rho/N}\mathbf{h}_i$ from the residual. Expanding this single change gives its exact objective improvement:

$$f(\mathbf{x}) - f(\mathbf{x}^{(i)}) = 4\sqrt{\frac{\rho}{N}} \mathbf{h}_i^\top \mathbf{w} + \frac{8\rho}{N} \mathbf{h}_i^\top \sum_{j \in S} \mathbf{h}_j - \frac{4\rho}{N} \|\mathbf{h}_i\|_2^2. \quad (34)$$

Controlling every expression in (34) separately would be unnecessarily difficult. Instead, sum the gains offered by all wrong bits. The cross terms then collapse into one squared norm.

Lemma 6.2 (Exact total gain offered by the wrong bits). *For every nonempty error set S ,*

$$\begin{aligned} &\sum_{i \in S} [f(\mathbf{x}) - f(\mathbf{x}^{(i)})] \\ &= 4 \left[\frac{\rho}{N} \left(2 \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 - \sum_{i \in S} \|\mathbf{h}_i\|_2^2 \right) + \sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i \right]. \end{aligned} \quad (35)$$

Proof. Sum the individual identity (34) over $i \in S$. The noise terms sum directly. For the middle terms,

$$\sum_{i \in S} \mathbf{h}_i^\top \sum_{j \in S} \mathbf{h}_j = \left(\sum_{i \in S} \mathbf{h}_i \right)^\top \left(\sum_{j \in S} \mathbf{h}_j \right) = \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2.$$

Substitution gives (35). □

Lemma 6.3 (From total gain to one profitable wrong bit). *If the right side of (35) is at least $a|S|$, then some wrong bit $i \in S$ satisfies*

$$f(\mathbf{x}) - f(\mathbf{x}^{(i)}) \geq a.$$

The maximal available gain is therefore also at least a . If $a \geq 1/N$, the detector accepts a flip with gain at least a .

Proof. The left side of (35) is a sum of $|S|$ wrong-bit gains. At least one summand is at least their average. The detector maximizes the gain over all N bits, so its chosen gain cannot be smaller than that particular wrong-bit gain. $\square$

Exact-algebra stage complete. The two exact identities depend only on the length of $\sum_{i \in S} h_i$, the individual column lengths, and the noise projection onto that column sum. Sections 7 and 8 control these quantities uniformly over all relevant error sets. In the outer region they will produce a wrong-bit gain of at least $\rho/2$; in the inner region, at least $4\rho/(\log N)^{5/4}$. Section 9 will use the objective gap as a barrier and as the finite budget that limits the total number of flips.

7 The outer region: constant-factor bounds are enough

The rounded LMMSE estimate begins with at most $N/(\log N)^{1/4}$ wrong bits. Greedy descent need not reduce Hamming distance at every step, so we prove the landscape estimates on the larger ball extending to $4N/(\log N)^{1/4}$ wrong bits. The factor four is chosen later to create an objective barrier; it is not a statistical threshold.

We split this ball into two cardinality ranges. This section treats

$$\left\lceil \frac{N}{(\log N)^3} \right\rceil < |S| \leq \left\lceil \frac{4N}{(\log N)^{1/4}} \right\rceil.$$

We call this the *outer region*. The word “outer” refers only to Hamming distance from the transmitted word.

For a fixed number k of wrong bits, a binary vector is determined by choosing which k coordinates are wrong. Therefore the number of vectors at Hamming distance exactly k from the truth is

$$\binom{N}{k}.$$

For every $1 \leq k \leq N$, the standard counting inequality gives

$$\binom{N}{k} \leq \left(\frac{eN}{k} \right)^k, \quad \log \binom{N}{k} \leq k \log \frac{eN}{k}.$$

Here “the logarithm of the number of sets” means the logarithm of the integer $\binom{N}{k}$; it is not a logarithm applied to a set. The relevant sets are precisely the possible error sets S in the Hamming range under discussion.

In the outer region, $k > N/(\log N)^3$, so

$$\frac{N}{k} < (\log N)^3.$$

Consequently,

$$\log \binom{N}{k} \leq k \log \frac{eN}{k} \leq k \log(e(\log N)^3) = k(1 + 3 \log \log N).$$

Equivalently, the number of vectors at distance k is at most

$$\exp\{k(1 + 3 \log \log N)\}.$$

This is the origin of the expression $e^{O(k \log \log N)}$. There are at most N possible integer values of k , because $k \in \{1, \dots, N\}$. This elementary observation is what later produces a multiplicative factor N when we sum bounds over all cardinalities.

Two deliberately coarse estimates suffice here. First, the squared norm of the erroneous channel sum $\|\sum_{i \in S} \mathbf{h}_i\|_2^2$ is within fixed constant factors of its mean $N|S|$. For one fixed set this estimate fails with probability e^{-cN} , and there are only $\exp\{o(N)\}$ sets in the controlled ball. Second, for one fixed set the probability that its noise projection is too large is $e^{-c|S| \log N}$. There are only $e^{O(|S| \log \log N)}$ sets of that size in this region. Thus both union bounds have ample slack.

Why two concentration regimes are needed. The sharper channel estimate used in [Section 8](#) has relative error $(\log N)^{-5/4}$. Instead of merely requiring the squared channel-sum length to lie between fixed multiples of its mean $N|S|$, we require it to lie in

$$\left[1 - \frac{1}{(\log N)^{5/4}}, 1 + \frac{1}{(\log N)^{5/4}}\right] N|S|.$$

For one fixed set S , the probability of violating this interval is at most

$$2 \exp \left\{ -\frac{N}{8(\log N)^{5/2}} \right\}.$$

The positive quantity $N/[8(\log N)^{5/2}]$ appearing after the minus sign is commonly called the concentration exponent or failure exponent. We use the phrase only as shorthand for this displayed probability bound.

If the inner region ended at $N/(\log N)^a$, the logarithm of the number of relevant sets would be of order $N \log \log N / (\log N)^a$. The latter is negligible compared with the former exactly when $a > 5/2$. We use the simple nearby choice $a = 3$. The split is not intrinsic: any fixed $a > 5/2$ would work. The choice $a = 5/2$ misses by a $\log \log N$ factor; a value such as $a = 11/2$ would work but would make the sharp inner region unnecessarily small.

Lemma 7.1 (Uniform length of erroneous channel sums). *With probability tending to one, simultaneously for every nonempty set with*

$$|S| \leq \left\lceil \frac{4N}{(\log N)^{1/4}} \right\rceil,$$

we have

$$\frac{3}{4}N|S| \leq \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 \leq \frac{5}{4}N|S|. \quad (36)$$

In particular, $\|\mathbf{h}_i\|_2^2 \leq 5N/4$ for every column.

Proof. Fix one nonempty set S . Since the columns are independent standard Gaussians,

$$\sum_{i \in S} \mathbf{h}_i \sim \mathcal{N}(\mathbf{0}, |S| \mathbf{I}_N).$$

It follows that

$$\frac{1}{|S|} \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 \sim \chi_N^2.$$

The chi-square bound with relative error $1/4$ says that this one fixed set violates the interval $[3N/4, 5N/4]$ with probability at most $2e^{-N/128}$.

We now count all sets for which this bound must hold. Put

$$m = \left\lceil \frac{4N}{(\log N)^{1/4}} \right\rceil.$$

This abbreviation is used only in the present counting paragraph. Since $k \mapsto k \log(eN/k)$ is increasing for $k < N$, for every allowed k ,

$$\binom{N}{k} \leq \exp \left\{ m \log \left(\frac{eN}{m} \right) \right\}$$

for all sufficiently large N , when $m < N$. There are at most N allowed values of k . Therefore the total number of nonempty sets in the stated range is at most N times the preceding display. Taking its logarithm gives

$$O \left(\frac{N \log \log N}{(\log N)^{1/4}} \right) = o(N).$$

The union bound over all those sets is consequently at most

$$2 \exp \left\{ O \left(\frac{N \log \log N}{(\log N)^{1/4}} \right) - \frac{N}{128} \right\},$$

which tends to zero because $N \log \log N / (\log N)^{1/4} = o(N)$.

Finally take a singleton set $S = \{i\}$. Then $\sum_{j \in S} \mathbf{h}_j = \mathbf{h}_i$. The upper half of the same uniform bound gives

$$\|\mathbf{h}_i\|_2^2 \leq \frac{5}{4}N$$

simultaneously for every column i . This is the “individual-column statement” used later when bounding $\sum_{i \in S} \|\mathbf{h}_i\|_2^2$. $\square$

Lemma 7.2 (Uniform control of outer-region noise projections). *There is one event, depending only on $\mathbf{H}$ and $\mathbf{w}$, whose probability tends to one and on which the following inequality holds for every $\rho \geq 2 \log N$ and every set in the outer region:*

$$\left| \sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i \right| \leq \frac{\rho |S|}{8}. \quad (37)$$

Proof. Condition on a channel for which every erroneous column sum has the length asserted in the preceding lemma. Fix one set S in the outer region. The vector $\sum_{i \in S} \mathbf{h}_i$ is now fixed, while $\mathbf{w}$ is still standard Gaussian. Therefore

$$\sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i$$

is a one-dimensional centered Gaussian. Its variance is

$$\frac{\rho}{N} \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2,$$

which is at most

$$\frac{\rho}{N} \cdot \frac{5}{4} N |S| = \frac{5}{4} \rho |S|.$$

Thus its typical magnitude is of order $\sqrt{\rho |S|}$. We need it to stay below $\rho |S|/8$, which is of order $\sqrt{\rho |S|}$ standard deviations from zero. Applying $\mathbb{P}\{|Z| \geq t\} \leq 2e^{-t^2/2}$ to this one fixed set gives

$$\mathbb{P} \left\{ \left| \sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i \right| > \frac{\rho |S|}{8} \mid \mathbf{H} \right\} \leq 2 \exp \left(-\frac{\rho |S|}{160} \right). \quad (38)$$

We now ask whether this estimate holds for every set simultaneously. At the smallest allowed SNR, $\rho = 2 \log N$, the failure exponent for one fixed set is $|S| \log N/80$. Here “one fixed set” refers to the conditional failure probability in the preceding display for one specified S , before we union-bound over the possible choices of S .

The number of sets having this cardinality is $\binom{N}{|S|}$. Because every outer-region cardinality satisfies $|S| > N/(\log N)^3$, the counting calculation given at the start of this section gives

$$\log \binom{N}{|S|} \leq |S| \log \frac{eN}{|S|} \leq |S| (1 + 3 \log \log N).$$

The phrase “for all sufficiently large N ” means that there exists a fixed finite integer N_0 such that the assertion holds for every $N \geq N_0$. It is needed here because the asymptotic fact $\log \log N = o(\log N)$ does not imply the displayed numerical inequality for every small or moderate N . For $N \geq N_0$,

$$1 + 3 \log \log N < \frac{\log N}{160}.$$

Therefore, for one fixed cardinality k in the outer region,

$$\begin{aligned} & \binom{N}{k} 2 \exp \left(-\frac{k \log N}{80} \right) \\ & \leq 2 \exp \left\{ k(1 + 3 \log \log N) - \frac{k \log N}{80} \right\} \\ & \leq 2 \exp \left(-\frac{k \log N}{160} \right). \end{aligned}$$

There are fewer than N possible cardinalities because k is an integer between one and N ; this does not mean that their number is $o(N)$. Summing the last bound over those cardinalities and using $k > N/(\log N)^3$ gives

$$2N \exp \left(-\frac{N \log N}{160(\log N)^3} \right) = o(1).$$

This tends to zero. Increasing ρ makes the required threshold larger in standard-deviation units: after dividing both sides of the desired bound by $\sqrt{\rho}$, its right-hand side grows as $\sqrt{\rho}$. Hence the

event proved at $\rho = 2 \log N$ automatically implies the corresponding event for every larger ρ on the same realization. This is the precise meaning of “uniformly over $\rho \geq 2 \log N$.”

If the channel-sum event from [Lemma 7.1](#) is denoted in words by “all coarse channel bounds hold,” the total failure probability in this lemma is at most

$$\mathbb{P}\{\text{some coarse channel bound fails}\} + 2N \exp\left(-\frac{N \log N}{160(\log N)^3}\right).$$

The first probability tends to zero by [Lemma 7.1](#), and the second displayed quantity tends to zero by direct inspection. This supplies an explicit probability bound rather than only the phrase “with probability tending to one.” $\square$

Lemma 7.3 (Excess cost and one-step progress in the outer region). *Suppose the following two statements hold simultaneously for every relevant error set: the squared channel-sum norm lies between $(3/4)N|S|$ and $(5/4)N|S|$, and the magnitude of the scaled noise projection is at most $\rho|S|/8$. The preceding two lemmas show that this joint event has probability tending to one. On this event, consider any binary vector whose error set satisfies*

$$\left\lceil \frac{N}{(\log N)^3} \right\rceil < |S| \leq \left\lfloor \frac{4N}{(\log N)^{1/4}} \right\rfloor.$$

Then

$$\frac{5}{2}\rho|S| \leq f(\mathbf{x}) - f(\mathbf{1}) \leq \frac{11}{2}\rho|S|, \quad (39)$$

and

$$\sum_{i \in S} [f(\mathbf{x}) - f(\mathbf{x}^{(i)})] \geq \frac{1}{2}\rho|S|. \quad (40)$$

Hence correcting at least one currently wrong bit decreases the cost by at least $\rho/2$.

Proof. The exact objective-gap identity (33) is

$$f(\mathbf{x}) - f(\mathbf{1}) = \frac{4\rho}{N} \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 + 4\sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i.$$

The channel-sum norm lies between $(3/4)N|S|$ and $(5/4)N|S|$, and the scaled noise projection has magnitude at most $\rho|S|/8$. The smallest possible coefficient of $\rho|S|$ is

$$4 \left(\frac{3}{4} - \frac{1}{8} \right) = \frac{5}{2},$$

and the largest possible coefficient is $4(5/4 + 1/8) = 11/2$. This proves the displayed objective bounds.

For progress, recall the exact identity for the sum of all gains offered by the currently wrong bits:

$$4 \left[\frac{\rho}{N} \left(2 \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 - \sum_{i \in S} \|\mathbf{h}_i\|_2^2 \right) + \sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i \right].$$

The gain identity contains three contributions inside its outer factor of four: twice the squared channel-sum norm, minus the sum of the individual column norms, plus the noise projection. To

obtain a lower bound, use the lower channel-sum bound, the upper singleton bound $\sum_{i \in S} \|h_i\|_2^2 \leq (5/4)N|S|$, and the most negative allowed noise projection. Substituting these three bounds directly into (35) gives

$$\begin{aligned} \sum_{i \in S} \Delta_i(\mathbf{x}) &\geq 4\rho|S| \left(2 \cdot \frac{3}{4} - \frac{5}{4} - \frac{1}{8} \right) \\ &= \frac{1}{2}\rho|S|. \end{aligned}$$

Thus the sum of the cost decreases obtained by correcting the $|S|$ currently wrong bits is at least $(\rho/2)|S|$. At least one of those gains is therefore at least their average, $\rho/2$. Since the greedy detector chooses the largest cost decrease among all N coordinates, its next accepted step decreases the objective by at least $\rho/2$. This statement guarantees objective progress; it does not claim that every bit selected by the greedy rule must currently be wrong. $\square$

Outer-region stage complete. Uniform constant-factor channel and noise bounds place every outer-region vector strictly above the truth and certify a wrong-bit gain of at least $\rho/2$. The lower objective bound at the outer boundary will become the barrier that keeps the descent path inside the controlled ball.

8 The inner region: sharp control near the truth

We now treat every nonempty error set satisfying

$$1 \leq |S| \leq \left\lceil \frac{N}{(\log N)^3} \right\rceil.$$

This is the *inner region*. It meets the outer region without a gap: the boundary cardinality belongs to the inner region, and the next larger cardinality belongs to the outer region.

The terminology can otherwise be misleading. The inner region contains the vectors *closer* to the truth and therefore has the smaller error cardinalities. Its concentration conclusion is tighter, not weaker: the channel norm must have relative error only $(\log N)^{-5/4}$. Such a tight statement can hold uniformly here precisely because the inner region contains far fewer candidate error sets than the full controlled ball.

The exact numbers of candidate binary vectors are

$$\sum_{k=1}^{\lceil N/(\log N)^3 \rceil} \binom{N}{k} \quad \text{in the inner region}$$

and

$$\sum_{k=\lceil N/(\log N)^3 \rceil + 1}^{\lceil 4N/(\log N)^{1/4} \rceil} \binom{N}{k} \quad \text{in the outer region.}$$

Their logarithms are bounded respectively by

$$O\left(\frac{N \log \log N}{(\log N)^3}\right) \quad \text{and} \quad O\left(\frac{N \log \log N}{(\log N)^{1/4}}\right).$$

The outer region has many more vectors; this is why it supports only the coarser uniform channel estimate.

Before proving concentration inequalities, we identify the exact Gaussian tail that matters. Suppose temporarily that $|S| = k$ and that the erroneous channel sum has its typical squared length,

$$\left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 \approx Nk.$$

The objective gap is

$$f(\mathbf{x}) - f(\mathbf{1}) = \frac{4\rho}{N} \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 + 4\sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i.$$

The positive signal term is therefore approximately $4\rho k$. For the noise to cancel it, one would need approximately

$$\mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i \leq -\sqrt{\rho N} k.$$

Conditional on the channel, the random variable on the left is Gaussian with standard deviation approximately $\sqrt{Nk}$. Dividing the required negative fluctuation by this standard deviation gives

$$\frac{-\sqrt{\rho N} k}{\sqrt{Nk}} = -\sqrt{\rho k}.$$

Thus the dangerous one-set probability is approximately

$$Q(\sqrt{\rho k}).$$

Here “dangerous one-set probability” means the probability, for one fixed error set S , that the negative noise projection is large enough to nearly cancel the positive signal contribution to the excess cost. If full cancellation occurred, that particular nontruth vector could have cost no larger than the truth. The rigorous lemma controls both signs of the noise because the later gain calculation also needs an upper-magnitude bound.

This calculation already reveals the constant two. If $\rho = c \log N$, the Gaussian tail is, up to its prefactor,

$$Q(\sqrt{ck \log N}) \approx \frac{N^{-ck/2}}{\sqrt{k \log N}}.$$

There are approximately $N^k/k!$ error sets of size k . Their total union-bound contribution is therefore approximately

$$\frac{N^{k(1-c/2)}}{k! \sqrt{k \log N}}.$$

For $c > 2$, a negative power of N remains. For $c = 2$, the powers of N cancel, but the Gaussian prefactor and $1/k!$ still make the sum tend to zero. For $c < 2$, the power of N has the wrong sign; already the one-bit contribution grows instead of vanishing. The rigorous proof below adds a small relative slack to this calculation but changes none of its leading terms.

Constant-factor concentration would perturb the exponent $ck/2$ by a constant factor and would therefore lose the boundary $c = 2$. We instead keep relative error only $(\log N)^{-5/4}$. For sets of size k , this changes the exponent by $9k/(\log N)^{1/4}$. The correction per unit k tends to zero; the

total correction need not. Equivalently, it contributes only an $e^{o(k)}$ factor, which the factorial in the subset count absorbs.

The achievability proof does not require a global comparison with all $2^N - 1$ nontruth words. It requires two properties only inside the controlled Hamming ball: every nontruth vector there must lie above the truth, and some correcting flip must lower its cost. In [Section 9](#), the objective barrier will prove that the algorithm never leaves this ball, so no landscape information outside it is needed.

Why vanishing relative error is needed, and why the exponent 5/4 is convenient. Suppose more generally that the relative-error level were $(\log N)^{-b}$. Preserving the sharp Gaussian exponent without a constant loss is automatic for $b > 1$. The endpoint $b = 1$ leaves only an e^{Ck} factor, which is still summable against $k!$. On the other hand, the fixed-set chi-square exponent would be $N/(\log N)^{2b}$. To union-bound over the inner sets, this must dominate their counting exponent $N \log \log N/(\log N)^3$, which holds when $b < 3/2$. Any fixed choice

$$1 \leq b < \frac{3}{2}$$

works. The exponent 5/4 is the midpoint of this admissible interval and keeps the later arithmetic simple; it is not a fundamental constant.

8.1 Establish uniform sharp channel geometry

The threshold calculation requires the squared channel-sum norm to remain within a vanishing relative error of its mean $N|S|$. We first prove that this sharper channel event holds simultaneously for every inner error set; the following subsection will then condition on it to control the noise.

Lemma 8.1 (Sharp concentration of the erroneous channel sum). *For all sufficiently large N , with probability tending to one, simultaneously for every nonempty set with*

$$|S| \leq \left\lceil \frac{N}{(\log N)^3} \right\rceil,$$

we have

$$\left(1 - \frac{1}{(\log N)^{5/4}}\right) N|S| \leq \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 \leq \left(1 + \frac{1}{(\log N)^{5/4}}\right) N|S|. \quad (41)$$

Proof. Fix a set of size $|S|$. The sum of its independent Gaussian columns has distribution $\mathcal{N}(\mathbf{0}, |S| \mathbf{I}_N)$. Therefore

$$\frac{1}{|S|} \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 \sim \chi_N^2.$$

Apply (10) with $\varepsilon = (\log N)^{-5/4}$. The failure probability for one fixed set is at most

$$2 \exp \left(-\frac{N}{8(\log N)^{5/2}} \right).$$

We next count the sets over which this fixed-set estimate must be union-bounded. For every integer $1 \leq k \leq \lceil N/(\log N)^3 \rceil$,

$$\binom{N}{k} \leq \exp \left\{ k \log \frac{eN}{k} \right\}.$$

The exponent $k \log(eN/k)$ increases with k for $k < N$. Hence every term in the sum over cardinalities is at most the value obtained at $k = \lceil N/(\log N)^3 \rceil$. There are at most $\lceil N/(\log N)^3 \rceil$ terms. Therefore the total number of nonempty inner-region error sets is at most

$$\left\lceil \frac{N}{(\log N)^3} \right\rceil \exp \left\{ \left\lceil \frac{N}{(\log N)^3} \right\rceil \log \left(\frac{eN}{\lceil N/(\log N)^3 \rceil} \right) \right\}.$$

Ignoring only ceiling corrections, the logarithm inside the exponential is

$$\log \left(\frac{eN}{N/(\log N)^3} \right) = 1 + 3 \log \log N.$$

It follows that the logarithm of the total number of inner-region sets is

$$O \left(\frac{N \log \log N}{(\log N)^3} \right).$$

The union bound is the number of sets multiplied by the fixed-set failure probability. In particular, the total failure probability is at most

$$2 \left\lceil \frac{N}{(\log N)^3} \right\rceil \exp \left\{ \left\lceil \frac{N}{(\log N)^3} \right\rceil \log \left(\frac{eN}{\lceil N/(\log N)^3 \rceil} \right) - \frac{N}{8(\log N)^{5/2}} \right\}.$$

To see directly that this expression tends to zero, divide its positive counting exponent by the negative fixed-set concentration exponent $N/[8(\log N)^{5/2}]$. The ratio is

$$O \left(\frac{\log \log N}{\sqrt{\log N}} \right) \rightarrow 0.$$

Thus the negative concentration exponent dominates the logarithm of the number of sets, and the total failure probability tends to zero. $\square$

8.2 Control all inner noise projections at the threshold scale

The preceding lemma fixes the conditional variance of every relevant noise projection up to the required relative error. We now use the resulting Gaussian tail at $\rho = 2 \log N$, followed by a union bound over the inner error sets. This is the step at which the leading constant 2 matters.

Lemma 8.2 (Sharp uniform bound on the noise projection). *For all sufficiently large N , there is one event, depending only on $\mathbf{H}$ and $\mathbf{w}$, whose probability tends to one and on which the following inequality holds for every $\rho \geq 2 \log N$ and every nonempty set satisfying*

$$|S| \leq \left\lceil \frac{N}{(\log N)^3} \right\rceil,$$

:

$$\left| \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i \right| \leq \left(1 - \frac{4}{(\log N)^{5/4}} \right) \sqrt{\rho N} |S|. \quad (42)$$

Proof. It is enough to prove the statement at $\rho = 2 \log N$, because its right-hand side increases with ρ . Thus the event at the boundary SNR is contained in the corresponding event at every larger SNR, which is again the precise meaning of uniformity over ρ .

Condition on a channel satisfying (41). For one fixed set, the projection is conditionally Gaussian:

$$w^\top \sum_{i \in S} h_i \mid H \sim \mathcal{N} \left(0, \left\| \sum_{i \in S} h_i \right\|_2^2 \right).$$

The forbidden magnitude in the lemma is

$$\left(1 - \frac{4}{(\log N)^{5/4}} \right) \sqrt{2N \log N} |S|.$$

The conditional standard deviation is the norm of the channel sum. The preceding lemma bounds that standard deviation by

$$\sqrt{\left(1 + \frac{1}{(\log N)^{5/4}} \right) N |S|}.$$

Dividing the forbidden magnitude by this upper bound produces

$$\sqrt{2|S| \log N \frac{(1 - 4/(\log N)^{5/4})^2}{1 + 1/(\log N)^{5/4}}}.$$

This is the rigorous version of the heuristic threshold $\sqrt{\rho|S|}$. The appearance of $\log N$ is not an additional assumption or a new concentration slack: we have simply substituted the boundary value $\rho = 2 \log N$, so $\sqrt{\rho|S|} = \sqrt{2|S| \log N}$. The extra fraction under the square root is the small relative slack created by the channel relative-error bound and by placing the permitted noise magnitude slightly below complete cancellation. The conditional probability of violating the desired two-sided bound is therefore at most

$$2Q \left(\sqrt{2|S| \log N \frac{(1 - 4/(\log N)^{5/4})^2}{1 + 1/(\log N)^{5/4}}} \right). \quad (43)$$

We now quantify that small relative slack. The fraction inside the square root satisfies

$$\begin{aligned} \frac{(1 - 4/(\log N)^{5/4})^2}{1 + 1/(\log N)^{5/4}} &= 1 - \frac{9}{(\log N)^{5/4}} + \frac{25/(\log N)^{5/2}}{1 + 1/(\log N)^{5/4}} \\ &\geq 1 - \frac{9}{(\log N)^{5/4}}. \end{aligned}$$

Thus the squared standardized threshold is at least $2|S| \log N [1 - 9/(\log N)^{5/4}]$. Mills' bound gives the following failure probability for this one fixed set:

$$\frac{2}{\sqrt{|S| \log N}} N^{-|S|(1 - 9/(\log N)^{5/4})}. \quad (44)$$

We now require the estimate for every error set simultaneously. Fix a cardinality k . There are $\binom{N}{k}$ sets of this size, so the union bound at this one cardinality is at most

$$\binom{N}{k} \frac{2}{\sqrt{k \log N}} N^{-k(1 - 9/(\log N)^{5/4})}$$

$$\begin{aligned}
&\leq \frac{N^k}{k!} \frac{2}{\sqrt{k \log N}} N^{-k} N^{9k/(\log N)^{5/4}} \\
&= \frac{2}{k! \sqrt{k \log N}} \exp\left(\frac{9k}{(\log N)^{1/4}}\right).
\end{aligned}$$

The first line is “number of sets” times “failure probability for one fixed set.” The second line uses $\binom{N}{k} \leq N^k/k!$ and separates the Gaussian power into its leading N^{-k} term and the relative-slack term. The factors N^k and N^{-k} cancel exactly. This is the only place where the constant two is used sharply.

Because $9/(\log N)^{1/4} \rightarrow 0$, for all sufficiently large N ,

$$\exp\left(\frac{9k}{(\log N)^{1/4}}\right) \leq 2^k.$$

Therefore the contribution from all sets of cardinality k is at most

$$\frac{2}{\sqrt{\log N}} \frac{2^k}{k! \sqrt{k}}.$$

Summing this expression over the actual inner cardinalities can only be smaller than summing it over every positive integer. Hence the complete conditional union bound is at most

$$\frac{2}{\sqrt{\log N}} \sum_{k=1}^{\infty} \frac{2^k}{k! \sqrt{k}} = O\left(\frac{1}{\sqrt{\log N}}\right).$$

The infinite series is a fixed finite constant, and the factor $1/\sqrt{\log N}$ tends to zero. Finally add the probability that the sharp channel-sum event fails. Thus an explicit upper bound for this lemma’s failure probability is

$$\mathbb{P}\{\text{some sharp channel-sum bound fails}\} + \frac{2}{\sqrt{\log N}} \sum_{k=1}^{\infty} \frac{2^k}{k! \sqrt{k}},$$

and both terms tend to zero. □

Where $2 \log N$ is used. For an error set of size k , the Gaussian tail supplies essentially N^{-k} , while the number of sets supplies essentially $N^k/k!$. The powers of N cancel. The Gaussian prefactor $1/\sqrt{k \log N}$ and the factorial $1/k!$ make the remaining sum vanish. This is the threshold-critical calculation.

8.3 Convert the uniform bounds into objective progress

We have now controlled the only random quantities appearing in the exact objective-gap and total-gain identities. Substituting those bounds into the identities yields the two deterministic landscape conclusions needed by the descent argument: positive excess cost and a quantified improving flip.

Lemma 8.3 (Excess cost and one-step progress in the inner region). *Suppose the following two statements hold simultaneously for every inner error set: the squared channel-sum norm lies between $[1 - (\log N)^{-5/4}]N|S|$ and $[1 + (\log N)^{-5/4}]N|S|$, and the magnitude of the unscaled noise projection is at most $[1 - 4(\log N)^{-5/4}]\sqrt{\rho N}|S|$. The preceding two lemmas show that this joint event has probability tending to one. On this event, for every error set satisfying*

$$1 \leq |S| \leq \left\lceil \frac{N}{(\log N)^3} \right\rceil,$$

$$\frac{12\rho}{(\log N)^{5/4}}|S| \leq f(\mathbf{x}) - f(\mathbf{1}) \leq 8\rho|S|, \quad (45)$$

and

$$\sum_{i \in S} [f(\mathbf{x}) - f(\mathbf{x}^{(i)})] \geq \frac{4\rho}{(\log N)^{5/4}}|S|. \quad (46)$$

Hence correcting at least one currently wrong bit decreases the cost by at least $4\rho / (\log N)^{5/4}$.

Proof. Recall that the objective gap equals

$$\frac{4\rho}{N} \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 + 4\sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i.$$

The squared channel-sum norm is at least $[1 - (\log N)^{-5/4}]N|S|$. The scaled noise projection can be as negative as

$$- \left(1 - \frac{4}{(\log N)^{5/4}} \right) \rho |S|.$$

Substituting these two worst-case values, the coefficient of $\rho|S|$ is

$$4 \left[1 - \frac{1}{(\log N)^{5/4}} - \left(1 - \frac{4}{(\log N)^{5/4}} \right) \right] = \frac{12}{(\log N)^{5/4}}.$$

Using the upper channel bound and the most positive allowed noise projection gives a coefficient no larger than 8. Hence every nontruth vector in the inner region lies strictly above the truth, and its excess cost is at most $8\rho|S|$.

For one-step progress, the exact total wrong-bit gain is

$$4 \left[\frac{\rho}{N} \left(2 \left\| \sum_{i \in S} \mathbf{h}_i \right\|_2^2 - \sum_{i \in S} \|\mathbf{h}_i\|_2^2 \right) + \sqrt{\frac{\rho}{N}} \mathbf{w}^\top \sum_{i \in S} \mathbf{h}_i \right].$$

Applying the sharp channel statement to singleton sets gives

$$\sum_{i \in S} \|\mathbf{h}_i\|_2^2 \leq \left(1 + \frac{1}{(\log N)^{5/4}} \right) N|S|.$$

To lower-bound the total gain, combine this upper bound with the lower bound on the squared channel-sum norm and the most negative permitted noise projection. Substitution into (35) gives

$$\sum_{i \in S} \Delta_i(\mathbf{x}) \geq 4\rho|S| \left[2 \left(1 - \frac{1}{(\log N)^{5/4}} \right) - \left(1 + \frac{1}{(\log N)^{5/4}} \right) - \left(1 - \frac{4}{(\log N)^{5/4}} \right) \right]$$

$$= \frac{4\rho|S|}{(\log N)^{5/4}}.$$

Thus the sum of the $|S|$ wrong-bit gains is at least

$$\frac{4\rho|S|}{(\log N)^{5/4}}.$$

At least one wrong bit therefore offers the average gain

$$\frac{4\rho}{(\log N)^{5/4}}.$$

At the boundary SNR $\rho = 2 \log N$, this is $8/(\log N)^{1/4}$. It may tend slowly to zero, but it remains much larger than the algorithm's acceptance floor $1/N$. No increase of the SNR is required. $\square$

The two-region landscape is complete. For every error cardinality from one through $4N/(\log N)^{1/4}$, exactly one of the two region statements applies. Every such nontruth vector lies strictly above the truth. Some wrong bit offers gain at least $\rho/2$ in the outer region and at least $4\rho/(\log N)^{5/4}$ in the inner region.

The second conclusion is stronger than merely saying that the truth has lower cost. It says that every controlled nontruth vector has an adjacent binary vector, obtained by correcting one currently wrong coordinate, with strictly smaller cost. Hence the controlled ball contains no spurious one-bit local minimum. The landscape may remain globally nonconvex, and the greedy path may temporarily flip a correct coordinate, but it cannot become trapped inside this basin.

These are uniform statements: one event asserts the inequalities for every eligible set S simultaneously. After that event has occurred, any set visited by the data-dependent greedy path is already covered. We never condition on which path the algorithm chooses, and independence between the path and the landscape is neither asserted nor needed.

9 The deterministic descent argument

The probabilistic work is now complete. Fix a realization on which the warm-start statement and all four uniform geometric statements hold. From this point onward, the argument is deterministic. Written out, the assumptions are:

1. For every outer error set, the squared channel-sum norm lies between $(3/4)N|S|$ and $(5/4)N|S|$.
2. For every outer error set, the scaled noise projection has magnitude at most $\rho|S|/8$.
3. For every inner error set, the squared channel-sum norm lies within the relative factor $1 \pm (\log N)^{-5/4}$ of $N|S|$.
4. For every inner error set, the unscaled noise projection has magnitude at most

$$\left[1 - \frac{4}{(\log N)^{5/4}}\right] \sqrt{\rho N}|S|.$$

The probability sections prove that these four statements and the LMMSE warm start hold together with probability tending to one. We now freeze any realization on which they all hold and show that the actual steepest-descent path reaches the truth before either stopping rule can activate.

The proof uses the cost f as a Lyapunov function. Hamming distance may move in either direction, but every accepted flip strictly decreases f . The uniform geometric events give three deterministic facts at every point in the controlled ball: every nontruth point has cost above the truth; the outer boundary has cost higher than the initialization; and some bit always offers a quantified decrease. Confinement and termination follow from these three comparisons.

The *cost budget* at a point $\mathbf{x}$ is simply its excess cost above the truth,

$$f(\mathbf{x}) - f(\mathbf{1}).$$

Because every controlled nontruth point has cost above $f(\mathbf{1})$, a downhill path starting from $\mathbf{x}$ can decrease its cost by at most this amount before reaching the truth's cost level. This is why an upper bound on excess cost, divided by a lower bound on the decrease per step, bounds the number of remaining steps.

Proposition 9.1 (Confinement and termination). *For all sufficiently large N and $\rho \geq 2 \log N$, if*

$$d_H(\mathbf{x}^{\text{init}}, \mathbf{1}) \leq \left\lceil \frac{N}{(\log N)^{1/4}} \right\rceil$$

and the conclusions of Lemmas 7.1, 7.2, 8.1 and 8.2 hold, then the detector reaches $\mathbf{1}$ before accepting $M_N = 4NL_N$ flips, with L_N as defined in Stage 2.

Proof. If $\mathbf{x}^{\text{init}} = \mathbf{1}$, the detector is already at the truth. The inner-region excess-cost lower bound, applied to every singleton error set, says that every one-bit neighbor has strictly larger cost. Hence every flip from the truth would increase the cost, and the detector stops. We henceforth assume that the initialization is not the truth.

Step 1: bound the starting excess cost. It does not matter for correctness whether the initialization begins in the inner or outer region; both are contained in the controlled ball. The distinction mattered only in the preceding probability proof, where the two regions required different estimates. The deterministic argument uses their common conclusions. In the inner region we proved $f(\mathbf{x}) - f(\mathbf{1}) \leq 8\rho|S|$. In the outer region we proved the smaller upper bound $f(\mathbf{x}) - f(\mathbf{1}) \leq (11/2)\rho|S|$. Since the warm start has at most $\lceil N/(\log N)^{1/4} \rceil$ errors, either case gives

$$f(\mathbf{x}^{\text{init}}) - f(\mathbf{1}) \leq 8\rho \left\lceil \frac{N}{(\log N)^{1/4}} \right\rceil. \quad (47)$$

Step 2: the decreasing-cost path cannot leave the controlled ball. Suppose, for the sake of the argument, that the path eventually exited the controlled ball. A bit flip changes Hamming distance by exactly one, so before exiting it would have to visit a vector at distance $\lceil 4N/(\log N)^{1/4} \rceil$. The outer-region lower bound says that the excess cost of every vector on this boundary is at least

$$f(\mathbf{x}) - f(\mathbf{1}) \geq \frac{5}{2}\rho \left\lceil \frac{4N}{(\log N)^{1/4}} \right\rceil. \quad (48)$$

This is a lower bound on $f(\mathbf{x}) - f(\mathbf{1})$, not on Hamming distance. To retain the ceiling terms explicitly, put $a_N = N/(\log N)^{1/4}$. The boundary height is at least

$$\frac{5}{2}\rho \lceil 4a_N \rceil \geq 10\rho a_N,$$

whereas (47) is at most

$$8\rho \lceil a_N \rceil \leq 8\rho(a_N + 1).$$

Because $a_N > 4$ for all sufficiently large N , the boundary height is strictly larger than the starting height.

Every accepted flip lowers f . Therefore every later iterate has cost at most the initialization's cost. But every point through which an escaping single-bit path would have to pass has cost strictly larger than the initialization's cost. Such a point is unreachable by a decreasing-cost path. The proposed exit is impossible, and every nontruth iterate remains in one of the two controlled regions.

Step 3: a nontruth vector cannot be a stopping point. Every nontruth iterate is now known to lie in one of the two regions. If it is in the outer region, some wrong bit offers gain at least $\rho/2$. If it is in the inner region, some wrong bit offers gain at least $4\rho/(\log N)^{5/4}$. The algorithm chooses the largest gain over all bits, so its selected cost decrease is at least the applicable value. Since $\rho \geq 2 \log N$,

$$\frac{4\rho}{(\log N)^{5/4}} \geq \frac{8}{(\log N)^{1/4}} > \frac{1}{N}$$

for all sufficiently large N . The detector's ordinary stopping rule is: stop if no bit offers a cost decrease of at least $1/N$. At every nontruth point, the displayed lower bound provides such a bit. Therefore the $1/N$ rule cannot stop the algorithm at a nontruth vector. Equivalently, on this good landscape there is no one-bit local minimum other than the truth anywhere in the controlled ball.

Step 4: recovery occurs within the allowed number of flips. At every confined nontruth point, the outer-region gain is at least $\rho/2$, while the inner-region gain is at least $4\rho/(\log N)^{5/4}$. For all sufficiently large N ,

$$\frac{\rho}{2} \geq \frac{4\rho}{(\log N)^{5/4}}.$$

Thus, regardless of which region contains the current iterate, every accepted flip whose starting point is not **1** decreases the objective by at least

$$g_N := \frac{4\rho}{(\log N)^{5/4}}.$$

This is a local abbreviation for the proof, not a quantity computed by the detector.

Suppose that T accepted flips occur without the path reaching **1**. Step 2 confines the endpoint x_T to the controlled ball. Since it is still a nontruth point, the uniform landscape lower bound gives $f(x_T) > f(\mathbf{1})$. Summing the decrease over the T flips and using (47) therefore yields

$$\begin{aligned} 0 < f(x_T) - f(\mathbf{1}) &\leq f(x^{\text{init}}) - f(\mathbf{1}) - Tg_N \\ &\leq 8\rho \left\lceil \frac{N}{(\log N)^{1/4}} \right\rceil - T \frac{4\rho}{(\log N)^{5/4}}. \end{aligned}$$

Consequently,

$$\begin{aligned} T &< 2(\log N)^{5/4} \left\lceil \frac{N}{(\log N)^{1/4}} \right\rceil \\ &\leq 2N \log N + 2(\log N)^{5/4} < 4N \log N \end{aligned}$$

for all sufficiently large N .

By Step 3, the algorithm continues to accept flips at every nontruth iterate. If it failed to reach $\mathbf{1}$ by the cap, then $T = M_N$ accepted flips would occur without a hit, contradicting $T < 4N \log N < M_N$. Hence a first accepted flip $R \geq 1$ whose post-flip iterate equals $\mathbf{1}$ exists. Applying the preceding no-hit inequality to the first $R - 1$ flips gives

$$R < 2(\log N)^{5/4} \left\lceil \frac{N}{(\log N)^{1/4}} \right\rceil + 1 < 4N \log N < M_N$$

for all sufficiently large N . Thus the detector cannot exhaust its allowed number of flips while the iterate is a nontruth vector.

Thus the detector reaches $\mathbf{1}$ strictly before the flip limit. At $\mathbf{1}$, the singleton inner-region lower bound says every one-bit neighbor has larger cost, so every flip has negative gain and the detector stops. $\square$

10 Proof of polynomial-time recovery and runtime

Proof of Theorem 2.1(a). By Proposition 5.4, the rounded LMMSE initialization satisfies

$$d_H(\mathbf{x}^{\text{init}}, \mathbf{x}^*) \leq \left\lceil \frac{N}{(\log N)^{1/4}} \right\rceil$$

with probability tending to one. The outer channel and noise conclusions follow from Lemmas 7.1 and 7.2; the corresponding inner conclusions follow from Lemmas 8.1 and 8.2. All these conclusions hold together with probability tending to one by the union bound; the events need not be independent.

On that intersection, Proposition 9.1 is deterministic and gives exact recovery. Column-sign symmetry transfers the conclusion from $\mathbf{x}^* = \mathbf{1}$ to every deterministic transmitted vector, with the same failure probability. Uniformity over $\rho \geq 2 \log N$ follows from the boundary reductions already made in the proof: the LMMSE matrix decreases in the positive-semidefinite order as ρ increases, and the outer and inner noise events at $2 \log N$ imply their counterparts at every larger SNR. Thus the sum of the displayed failure bounds controls the supremum in (5) and tends to zero.

For the runtime, form $\mathbf{G} = \mathbf{H}^\top \mathbf{H}$ and solve the dense ridge system in (4); these operations cost $O(N^3)$. Put $a = \sqrt{\rho/N}$, maintain the residual $\mathbf{r} = \mathbf{y} - a\mathbf{H}\mathbf{x}$, and set $\mathbf{q} = \mathbf{H}^\top \mathbf{r}$. Direct expansion gives every gain as

$$\Delta_j(\mathbf{x}) = -4ax_jq_j - 4a^2G_{jj}.$$

After flipping coordinate i , if $s = x_i$ denotes its pre-flip sign,

$$\mathbf{q}' = \mathbf{q} + 2as\mathbf{G}_{:,i}.$$

Thus one Gram-matrix column updates all N gains, and one scan and update costs $O(N)$. The deterministic flip limit $M_N = 4NL_N = O(N \log N)$ therefore makes the descent stage cost $O(N^2 \log N)$. The dense LMMSE solve still dominates, so the overall runtime is $O(N^3)$. These are unit-cost exact-real arithmetic operations in the model stated in Section 2; no finite-precision or bit-complexity conclusion is asserted. $\square$

Achievability proof complete. On the common high-probability event, rounded LMMSE starts inside the controlled ball, the objective barrier prevents escape, every nontruth iterate offers a decrease larger than the stopping threshold, and the total objective budget permits fewer than M_N accepted flips before recovery. Thus the detector returns $\mathbf{x}^*$ with uniformly vanishing failure probability and worst-case arithmetic cost $O(N^3)$.

11 Maximum likelihood fails below the first-order boundary

The recovery proof is complete. We now establish the converse in [Theorem 2.1\(b\)](#). The obstruction is deliberately local: if even one Hamming-distance-one neighbor has smaller objective value than the transmitted vector, then the transmitted vector cannot be an ML minimizer. The proof proceeds through four conceptual reductions, implemented in six lemmas:

$$\begin{aligned} \text{one-bit objective gap} &\implies \text{one short Gaussian interval} \implies \text{one bad column} \\ &\implies \text{at least one of the } N \text{ columns is bad.} \end{aligned}$$

As before, column-sign symmetry reduces the proof to $\mathbf{x}^* = \mathbf{1}$. Let $\mathbf{h}_i$ denote column i of $\mathbf{H}$.

11.1 Reduce ML failure to a one-bit inequality

It is enough to identify one one-bit neighbor whose objective is below the truth. The first lemma writes that comparison as the sign of a single scalar.

Lemma 11.1 (Exact cost of one wrong flip). *Let $\mathbf{1}^{(i)}$ be obtained from $\mathbf{1}$ by flipping coordinate i . Then*

$$f(\mathbf{1}^{(i)}) - f(\mathbf{1}) = 4T_i(\rho), \quad T_i(\rho) = \frac{\rho}{N} \|\mathbf{h}_i\|_2^2 + \sqrt{\frac{\rho}{N}} \mathbf{h}_i^\top \mathbf{w}. \quad (49)$$

Thus $T_i(\rho) < 0$ certifies that the truth is not an ML minimizer.

Proof. At the truth, the residual is $\mathbf{w}$. After flipping bit i , it is $\mathbf{w} + 2\sqrt{\rho/N} \mathbf{h}_i$. Expanding the difference of the two squared norms gives

$$\|\mathbf{w} + 2\sqrt{\rho/N} \mathbf{h}_i\|_2^2 - \|\mathbf{w}\|_2^2 = 4\frac{\rho}{N} \|\mathbf{h}_i\|_2^2 + 4\sqrt{\frac{\rho}{N}} \mathbf{h}_i^\top \mathbf{w}.$$

□

It is enough to work at the largest SNR in the converse range. The next one-line monotonicity statement explains why.

Lemma 11.2 (A bad bit remains bad at smaller SNR). *For fixed $(\mathbf{H}, \mathbf{w})$, if $T_i(\rho_0) < 0$, then $T_i(\rho) < 0$ for every $0 < \rho \leq \rho_0$.*

Proof. Divide the displayed expression for $T_i(\rho)$ in (49) by $\sqrt{\rho} > 0$:

$$\frac{T_i(\rho)}{\sqrt{\rho}} = \sqrt{\rho} \frac{\|h_i\|_2^2}{N} + \frac{h_i^\top w}{\sqrt{N}}.$$

The right side is nondecreasing in ρ . □

For the rest of the converse, set

$$\rho_0 = 2 \log N - \log \log N - s_N. \quad (50)$$

Because $s_N = o(\log N)$, this value is positive for all sufficiently large N , and $\rho_0 \sim 2 \log N$.

11.2 Construct a lower-cost neighbor from a Gaussian interval

Condition temporarily on a fixed noise vector. For a standard Gaussian column h independent of w , its signed projection onto the noise direction is

$$g = h^\top \frac{w}{\|w\|_2}. \quad (51)$$

Conditional on w , rotational invariance gives $g \sim \mathcal{N}(0, 1)$. We use the relative-error scale $N^{-1/4}$: it tends to zero, its chi-square exponent $N(N^{-1/4})^2 = \sqrt{N}$ diverges, and $\rho_0 N^{-1/4} \rightarrow 0$. These are precisely the three properties needed below.

Lemma 11.3 (A sufficient Gaussian interval). *Suppose*

$$\|w\|_2^2 \geq (1 - N^{-1/4})N, \quad \|h\|_2^2 \leq (1 + N^{-1/4})N. \quad (52)$$

If

$$-(1 + 5N^{-1/4})\sqrt{\rho_0} - \frac{1}{\sqrt{\rho_0}} \leq g \leq -(1 + 5N^{-1/4})\sqrt{\rho_0}, \quad (53)$$

then the corresponding $T_i(\rho_0)$ is negative for all sufficiently large N .

Proof. Using $h^\top w = g\|w\|_2$, the inequality $T_i(\rho_0) < 0$ is exactly

$$g < -\sqrt{\frac{\rho_0}{N}} \frac{\|h\|_2^2}{\|w\|_2}. \quad (54)$$

Under (52), the magnitude of the threshold on the right is at most

$$\sqrt{\rho_0} \frac{1 + N^{-1/4}}{\sqrt{1 - N^{-1/4}}} \leq (1 + 4N^{-1/4})\sqrt{\rho_0}$$

for all sufficiently large N . Every g in (53) is no greater than $-(1 + 5N^{-1/4})\sqrt{\rho_0}$, which is strictly below the required threshold. □

The interval has width $1/\sqrt{\rho_0}$. Keeping that width is what produces the $-\log \log N$ correction.

Lemma 11.4 (Probability of the sufficient interval). *For a standard Gaussian g ,*

$$\mathbb{P}\{g \text{ satisfies (53)}\} \geq (e^{-1} + o(1)) \frac{\phi(\sqrt{\rho_0})}{\sqrt{\rho_0}}. \quad (55)$$

Proof. On the negative half-line, the smallest density in the interval occurs at its left endpoint. Hence interval length times minimum density gives

$$\mathbb{P}\{g \text{ satisfies (53)}\} \geq \frac{1}{\sqrt{\rho_0}} \phi\left((1 + 5N^{-1/4})\sqrt{\rho_0} + \frac{1}{\sqrt{\rho_0}}\right).$$

The square of the displayed argument minus ρ_0 equals

$$10N^{-1/4}\rho_0 + 25N^{-1/2}\rho_0 + 2 + 10N^{-1/4} + \frac{1}{\rho_0} = 2 + o(1).$$

Since $\phi(t) = (2\pi)^{-1/2}e^{-t^2/2}$, the density ratio is $e^{-1} + o(1)$, proving the claim. $\square$

The projection g and the column length are not independent. The next lemma avoids any independence claim by subtracting the probability of an atypically long column.

Lemma 11.5 (One-column probability). *For every fixed w satisfying the first inequality in (52),*

$$\mathbb{P}\{T_i(\rho_0) < 0 \mid w\} \geq (e^{-1} + o(1)) \frac{\phi(\sqrt{\rho_0})}{\sqrt{\rho_0}}, \quad (56)$$

uniformly over all such w . Here only the independent column h_i remains random under the conditional probability.

Proof. Let A be the event that the column-length inequality in (52) holds, and let C be the Gaussian interval event (53). The preceding sufficient-interval lemma gives $A \cap C \subseteq \{T_i(\rho_0) < 0\}$. For arbitrary events, $\mathbb{P}(A \cap C) \geq \mathbb{P}(C) - \mathbb{P}(A^c)$. The chi-square bound (10) therefore gives

$$\mathbb{P}\{T_i(\rho_0) < 0 \mid w\} \geq (e^{-1} + o(1)) \frac{\phi(\sqrt{\rho_0})}{\sqrt{\rho_0}} - 2e^{-\sqrt{N}/8}.$$

Because $\rho_0 < 2 \log N$,

$$\frac{\phi(\sqrt{\rho_0})}{\sqrt{\rho_0}} \geq \frac{1}{N\sqrt{4\pi \log N}}.$$

The stretched-exponentially small final term is therefore negligible relative to the first, which proves (56). $\square$

Where the $\log \log N$ correction comes from. One column is bad with probability on the scale

$$\frac{\phi(\sqrt{\rho})}{\sqrt{\rho}} = \frac{e^{-\rho/2}}{\sqrt{2\pi\rho}}.$$

The factor $e^{-\rho/2}$ produces $2 \log N$. The interval has width $1/\sqrt{\rho}$; after multiplication by N , this factor produces the additional $-\log \log N$ term. Dropping the prefactor would lose the lower-order boundary proved here.

11.3 Use conditional independence across the columns

Conditional on the common noise vector, different one-bit tests use different independent channel columns. This turns the one-column estimate into N independent opportunities for a lower-cost neighbor. It remains to bound the probability that none of them occurs and then evaluate the result at ρ_0 .

Lemma 11.6 (At least one lower-cost neighbor appears). *For every w satisfying the first inequality in (52),*

$$\mathbb{P}\{T_i(\rho_0) \geq 0 \text{ for every } i \mid w\} \leq \exp\left\{-(e^{-1} + o(1))N \frac{\phi(\sqrt{\rho_0})}{\sqrt{\rho_0}}\right\}. \quad (57)$$

Proof. Given w , the N events depend on independent and identically distributed columns. If the probability that one column gives $T_i(\rho_0) < 0$ is p , the probability that no column does so is $(1 - p)^N \leq e^{-Np}$. Apply Lemma 11.5. $\square$

Proof of Theorem 2.1(b). First take $x^* = \mathbf{1}$ and $\rho = \rho_0$. Split according to whether $\|w\|_2^2 \geq (1 - N^{-1/4})N$. The chi-square bound and Lemma 11.6 give

$$\begin{aligned} & \mathbb{P}\{T_i(\rho_0) \geq 0 \text{ for every } i\} \\ & \leq 2e^{-\sqrt{N}/8} + \exp\left\{-(e^{-1} + o(1))N \frac{\phi(\sqrt{\rho_0})}{\sqrt{\rho_0}}\right\}. \end{aligned} \quad (58)$$

It remains to evaluate the exponent. Substituting (50) gives

$$\begin{aligned} N \frac{\phi(\sqrt{\rho_0})}{\sqrt{\rho_0}} &= \frac{N}{\sqrt{2\pi\rho_0}} e^{-\rho_0/2} \\ &= \frac{1 + o(1)}{\sqrt{4\pi}} e^{s_N/2} \rightarrow \infty. \end{aligned}$$

Both terms in (58) vanish. With probability tending to one, some $T_i(\rho_0) < 0$, and Lemma 11.1 then gives a strictly lower-cost one-bit neighbor.

In particular,

$$\{\mathbf{1} \in \arg \min_x f(x)\} \subseteq \bigcap_{i=1}^N \{T_i(\rho_0) \geq 0\},$$

so the preceding probability bound is exactly sufficient for the claimed ML-minimizer failure.

By Lemma 11.2, the same coupled event implies failure at every $0 < \rho \leq \rho_0$. Column-sign symmetry gives the same probability for every deterministic transmitted word. Consequently the bound controls the supremum in (6), proving the claimed uniform ML failure. $\square$

Converse proof complete. At or above $2 \log N$, the explicit two-stage detector achieves exact block recovery with uniformly vanishing error probability. At or below $2 \log N - \log \log N - s_N$, even one-bit competitors prevent ML recovery with uniformly high probability. The two statements meet at the sharp first-order constant 2; the remaining lower-order window is not resolved here.

A Decision-theoretic consequence of the converse

First-order ML achievability was already identified in Hansen et al. [2009] and stated with a diverging additive slack in Hassibi et al. [2014, Lemma IV.2]. The new constructive content of this paper is therefore Theorem 2.1(a), not another proof of the known first-order ML upper bound. This appendix records the standard decision-theoretic implication needed only if “statistical threshold” is read in the minimax sense. It is not used anywhere in the constructive proof.

Corollary A.1 (Minimax consequence). *Write $\mathbb{P}_{x^*,\rho}$ for probability under (1) with the displayed deterministic word and SNR. For the sequence in Theorem 2.1(b), put $\rho_{0,N} = 2 \log N - \log \log N - s_N$ and define the minimax risk at a fixed SNR by*

$$R_N^*(\rho) = \inf_{\delta_N} \sup_{x^* \in \{\pm 1\}^N} \mathbb{P}_{x^*,\rho} \{\delta_N(\mathbf{H}, \mathbf{y}, \rho) \neq x^*\},$$

where the infimum includes randomized detectors. For a randomized detector, the displayed probability also averages over its independent internal randomization. Then

$$\lim_{N \rightarrow \infty} \inf_{0 < \rho \leq \rho_{0,N}} R_N^*(\rho) = 1. \quad (59)$$

Thus the statistical and polynomial-time exact-recovery thresholds have the same first-order coefficient 2 in the unit-cost exact-real model.

Proof. Give $X \in \{\pm 1\}^N$ the uniform prior, independently of $(\mathbf{H}, w)$. Conditional on $\mathbf{H}$,

$$p(\mathbf{y} \mid \mathbf{H}, X = \mathbf{x}) \propto \exp\{-f(\mathbf{x})/2\}.$$

Consequently, least-squares ML is a MAP rule and minimizes the uniform-prior average block-error probability. Define

$$\eta_N = \sup_{\substack{x \in \{\pm 1\}^N \\ 0 < \rho \leq \rho_{0,N}}} \mathbb{P}_{x,\rho} \{x \in \arg \min_z f(z)\}.$$

By Theorem 2.1(b), $\eta_N \rightarrow 0$. At any fixed SNR in the lower regime, the uniform-prior average success probability of MAP is at most η_N . Bayes optimality implies that the average success probability of every detector is also at most η_N . Hence, for every detector,

$$\sup_x \mathbb{P}_{x,\rho} \{\delta_N \neq x\} \geq 2^{-N} \sum_x \mathbb{P}_{x,\rho} \{\delta_N \neq x\} \geq 1 - \eta_N.$$

The bound is uniform for $0 < \rho \leq \rho_{0,N}$, proving (59). $\square$

Remark A.2 (The ML upper endpoint). The preceding corollary concerns the lower, minimax side only. For completeness, Theorem 2.1(a) also implies

$$\lim_{N \rightarrow \infty} \sup_{\substack{x^* \in \{\pm 1\}^N \\ \rho \geq 2 \log N}} \mathbb{P}_{x^*,\rho} \{\hat{x}_{\text{ML}} \neq x^*\} = 0$$

for a measurable least-squares ML selector. This is a risk comparison, not an eventwise implication from successful local descent. Indeed, under the uniform prior used above, MAP optimality makes the average ML error no larger than the constructive detector’s average error. Objective ties have probability zero: conditional on an almost surely nonsingular $\mathbf{H}$, every pairwise cost difference is a nondegenerate affine Gaussian function of w . Finally, column-sign symmetry makes the ML error probability identical for every deterministic transmitted word, so its average and worst-word risks coincide. This observation is included only for completeness; first-order ML achievability was already known from the works cited at the start of this appendix.

B Equivalent formulations and adjacent literatures

The observation law itself has several exact non-MIMO descriptions. Writing $A = H/\sqrt{N}$, it is the load-one uncoded binary-CDMA model with user energy ρ , and it is the $B = 1$, $L = n = N$ signed Gaussian superposition construction [Joseph and Barron, 2012; Liu et al., 2024]. Moreover, if $b^* = (x^* + \mathbf{1})/2$, then

$$y + \sqrt{\rho/N} H\mathbf{1} = H(2\sqrt{\rho/N} b^*) + w, \quad b^* \in \{0, 1\}^N,$$

so the same data also form a dense binary Gaussian-regression problem. These are identities of the finite-dimensional joint observation law, not merely analogies between codeword or covariance distributions.

The closest coding and CDMA results do not provide the block-recovery theorem needed here. Joseph and Barron [2012] identify the $B = 1$ signed construction as a Gaussian generator-matrix code, but their reliability theorem assumes $B = L^n$ with a positive section-size exponent. Thus it does not include $B = 1$; at that endpoint their least-squares decoder searches all sign vectors, and their outer-code conversion to block reliability changes the message set. Likewise, the nonasymptotic AMP theorem of Rush and Venkataramanan [2019] controls a positive section-error fraction for sparse superposition codes at fixed rate with growing section size; it cannot be specialized by setting the section alphabet to two to obtain exact recovery here. The uncoded specialization of Liu et al. [2024] has exactly the present load-one CDMA law, but its state-evolution theorem characterizes fixed-iteration empirical user- and bit-error rates at fixed system parameters. It does not control the event that all users are correct when the energy grows as $\rho_N \asymp \log N$.

The regression identity gives another close comparison. For balanced words, which are included in our uniform theorem, it has $p = n = N$, support size $s = N/2$, and nonzero coefficient $a_N = 2\sqrt{\rho/N}$. The polynomial selector of Ndaoud and Tsybakov [2020] splits the observations into estimation and thresholding samples; its two stated sufficient sample-size conditions cannot both be met within $n = N$ under this dense specialization. The iterative compressed-sensing theorem of Gao and Zhang [2022] avoids that split, but assumes both $\limsup s/p < 1/2$ and $s(\log p)^4 = o(n)$, excluding $s = p/2$ and $n = p = N$. The polynomial-time correlation guarantee of Mazumdar and Sangwan [2025] specializes to an $N \log N$ observation scale in the dense case, while its sharper linear-regression likelihood result uses exhaustive fixed-weight decoding. Finally, the lattice method of Gamarnik et al. [2021] gives bit-polynomial exact recovery for dense rational regression, but at $n = p = N$ its proved additive-noise tolerance is exponentially small. In the corresponding variance-one-design normalization, the noise standard deviation here is $\sqrt{N/\rho} = \Theta(\sqrt{N/\log N})$.

There is also an exact conditional negative-Wishart statistic. Put $u = x^*/\sqrt{N}$, let $U_y \in \mathbb{R}^{(N-1) \times N}$ have orthonormal rows spanning $y^\perp$, and form $B = U_y H$. Gaussian conditioning gives, conditional on y , independent rows

$$B_{k,:} \sim \mathcal{N}\left(0, I_N - \frac{\rho}{1+\rho} uu^\top\right), \quad k = 1, \dots, N-1.$$

Thus B is a negative-spiked Rademacher Wishart instance with $p = N$, $n_{\text{samp}} = N-1$, and $\beta_N = -\rho/(1+\rho)$. At $\rho \asymp 2 \log N$, its residual variance in the planted direction is $1 + \beta_N = (1+\rho)^{-1} \asymp (2 \log N)^{-1}$. The statistic determines the word only up to a global sign, which the original observation can orient by comparing the two residual costs. This is a one-way statistic, not an equivalence of experiments, because the projection may discard information.

The neighboring Wishart literature treats different tasks or different parts of the hard edge. [Bandeira et al. \[2020\]](#) study detection and constrained-PCA certification, including low-degree evidence rather than a general computational-hardness theorem; they do not give a spike recovery algorithm for the triangular sequence above. [Zadik et al. \[2022\]](#) use lattice basis reduction to recover a planted hypercube at the exactly singular endpoint $\beta = -1$, where the planted vector is an exact null direction. Their guarantee relies on that noiseless algebraic structure. [Venkat \[2023\]](#) gives lattice-based discrepancy certificates and negative-Wishart detection in a much more nearly singular regime; at square aspect its guarantee requires exponentially small residual variance and does not recover the spike. None therefore yields recovery at $\beta_N = -1 + \Theta(1/\log N)$.

Accordingly, we do not claim novelty for the observation law, its coding, CDMA, or regression interpretations, the conditional Wishart statistic, or the local-search architecture itself. To our knowledge, no prior published theorem gives a detector that, using polynomially many unit-cost exact-real operations, has vanishing whole-vector error for the i.i.d. square signed-Gaussian code uniformly over every deterministic transmitted word and every $\rho \geq 2 \log N$.

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